

WHERE THE SECONDS GO: BOUNDING MIXTURE-OF-EXPERTS DECODE FROM HOST MEMORY, AND MEASURING WHAT FORESIGHT IS WORTH

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ABSTRACT

A Mixture-of-Experts model too large for one consumer GPU keeps most of its experts in host memory, and decoding one token at a time then waits on host reads. We ask how fast such decoding could be on a given machine, where the rest of the time goes, and what knowing the future routing is worth. No cache of C experts per layer reads fewer experts from host memory than Belady’s MIN with bypass, so MIN’s reads at the machine’s probed read rate bound every system that executes the exact routing. Our llama.cpp expert cache, faster than two other systems at most configurations, runs at 31–54% of that bound’s speed at the smallest gpt-oss budget on 25 machines with consumer processors. Oracles built into the engine then remove parts of the rest of its time, the gap. At that budget, on the 13 machines whose PCIe link is fast, caching MIN’s set and reading each admitted expert once close almost none of the gap alone but 39% together; issuing the copies ahead closes another 15%, and most of what remains is the GPU’s own work waiting in series with the reads. Knowing the future thus closes about half of the gap, but only when what is cached and how it is read change together and the link is fast. On our workload, exact routing of the next token would recover at most a third of what full foresight saves in host reads; half of it needs the routing of about 0.65 C distinct experts per layer ahead across 9 models.

1 INTRODUCTION

Open-weight Mixture-of-Experts (MoE) models now carry 30–120 B parameters and activate a few billion per token; a desktop GPU has 24–32 GB. Running such a model for one user therefore means keeping most experts in host DRAM. Each decoded token reads its routed experts from GPU memory if they are resident, and otherwise from host memory: either the CPU runs the expert, or the expert is copied over PCIe into a GPU slot (Ivanov et al., 2021; Hoefler et al., 2021).

Many systems build on this: GPU expert caches (Eliseev & Mazur, 2023; Xue et al., 2024; Song et al., 2024a), CPU execution of misses (Kamahori et al., 2024; Chen et al., 2025a; Zhong et al., 2025), a per-miss choice between the two (Yang et al., 2026; Zhu et al., 2026) and expert prediction (Hwang et al., 2024; Zhu et al., 2025; Wang et al., 2026b). Each reports a speed-up over a baseline of its choosing. None says how fast the same machine could decode the model, or which part of the remaining time a better design could remove. Without that, a builder cannot tell whether to work on what is cached, how it is read, when it is read, or

how far ahead the routing must be predicted.

We answer these questions for batch-1 decode of gpt-oss-120b (OpenAI, 2025) and Qwen3-30B-A3B (Yang et al., 2025) on rented RTX 5090 machines. We derive a bound, build oracles into a real engine that remove one part of the gap at a time, and run them on many machines, with every prediction committed before its machine started. Table 1 lists the answers and the evidence behind each.

The headroom is large (Section 3). MIN’s host reads at the machine’s probed rate bound the time per token of any system with C slots per layer, and that read time is nearly reachable in a microbenchmark. Our cache, which leads FreeToken and llama.cpp at most configurations, runs at roughly a third to a half of the bound’s speed at the smallest gpt-oss budget. Scored against bounds at datasheet rates, published batch-1 systems reach a median 13.6% of theirs, and ours 27%.

What is cached and how it is read pay only together (Section 4). At the smallest gpt-oss budget, caching MIN’s set while still reading each admission twice closes almost none of the gap, and so does reading once while caching the deployed set; doing both closes 39%. Reading ahead closes another 15%, and the GPU’s own work in series with the reads is most of the rest (Fig. 2).

Foresight pays only if it is spent right (Section 5). Knowing the future routing helps when each admitted expert is

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Table 1. Our claims and the evidence behind each. *Registered*: predicted in a job script committed before the machine started (Appendix C); *exploratory*: found in the data it describes; *derived*: follows from measurements and definitions. *Held* is mostly on the point estimate; Table 8 counts the clauses that held with an interval.

Claim	Evidence	§
MIN’s reads at the probed rate bound any exact-routing cache with C slots per layer	derived	3
The bound’s read time is nearly reachable	registered; read-time thresholds held, an overhead clause failed (3 hosts)	3
Our cache leads FreeToken and llama.cpp at most configurations	registered and held on host B; host S’s registered bands failed	3
What is cached and how it is read pay only together at gpt-oss 11%	states measured in registered runs; split and link subset after the fact (15 machines)	4
Foresight pays when MIN’s set is read once	registered; held at most cells on three fast-link machines; link scope after the fact	5
MIN’s fewest-admission schedule gains on every machine that ran it stably	registered on 7, held; two losses on unsteady machines	5
The link-to-CPU ratio decides which read path pays	exploratory (19 machines)	5
Admitting less gains a little online	registered; partly held	5
Half of foresight’s value needs $\approx 0.65 C$ distinct experts ahead	exploratory (9 models)	6
A closed-form account of the time	exploratory; of four registered tests, three failed, one inconclusive	K

read once, when the schedule admits as rarely as optimality allows, and when the read path suits the machine: where the PCIe link reads at 0.32 of the CPU rate or less, copies belong in the background.

Foresight has a price (Section 6). Half of its value needs the routing of about $0.65 C$ distinct experts per layer ahead, on 9 models. On our workload, exact routing of the next token recovers at most a third of it, and no forecaster we tested comes close.

Section 7 turns these results into rules for system builders.

Contributions.

- **A bound on time per token** for exact-routing systems with C slots per layer, shown to be nearly reachable in its read time, and the distance of our cache and of published systems from it (Section 3).
- **A measured decomposition of the gap** into five parts, three of them removed one at a time by oracles inside a real engine on 15 machines (Section 4).
- **How foresight must be spent**: read once, with the fewest admissions, on the path the machine favours (Section 5).
- **A price on foresight** by horizon and accuracy, and a horizon rule in distinct experts across 9 models (Section 6).
- **An open record**: every prediction committed before the machine it concerns started, including the ones that failed, and every number generated by a script (Appendices C and O).

2 SETTING

Models and workload. gpt-oss-120b has 36 MoE layers of 128 experts, top-4, in MXFP4 (13.25 MB per expert). Qwen3-30B-A3B has 48 layers of 128 experts, top-8, in BF16 (9.44 MB per expert). Prompts are AIME-25 problems, and the cache carries over between problems, as in one user’s session. The system comparison decodes 256 tokens of each of the 30 problems through each system’s server. The engine experiments replay the model’s own greedy output, teacher-forced, so that every configuration sees the same routing. A *budget* is the share C/E of each layer’s experts that fit in GPU slots. The two smallest budgets of each model are *host-bound*: host reads, not the GPU, limit their speed. Most results are at the two host-bound gpt-oss budgets, 11% and 25%.

The engine. Our cache is a 3,500-line patch to llama.cpp (Gerganov & contributors, 2023–2026). Each MoE layer has C GPU slots. A *miss* is read from host memory in one of two ways: CPU helper threads run it, or it is *fetched* (copied into a slot over PCIe and run on the GPU in the same step). A per-machine table, computed from the machine’s bandwidth probe, sets how many of a layer’s misses are fetched. The *deployed* policy admits a miss into a slot when its decayed request count beats the lowest resident’s by a margin $\kappa=1$. The CPU serves the admitted miss and a background copy then fills its slot, so the expert is read from host memory twice (Fig. 1).

Terms. *MIN* is Belady’s optimum with bypass (Belady, 1966; Zhang et al., 2020): it *admits* a miss only if some

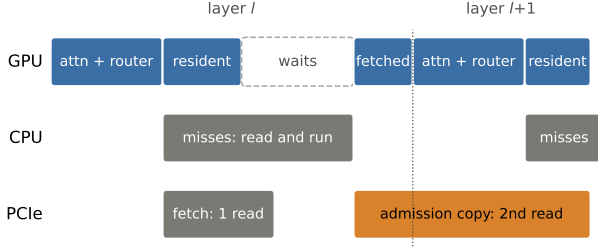


Figure 1. One decode step of the deployed cache, schematically. Blue: the GPU’s own kernels. Grey: host reads of the misses, which the step waits for. Orange: the background copy that fills an admitted expert’s slot, a second read of the same expert.

resident is needed later than the miss, which gives the most hits possible with C slots (Carlisle & Lloyd, 1995; Jain & Lin, 2016). Many schedules reach that optimum. The *greedy* one admits whenever it may; the *fewest-admission* one admits as rarely as any of them. An admitted expert is read *once*, copied into its slot before the GPU runs it, or *twice*, served by the CPU and copied in the background. It is read *late* if the copy waits for the router, and *ahead* if the copy is issued a few steps early. *Oracles* read the future routing from a trace recorded on the same machine. Table 2 lists every configuration; Appendix B maps the codes of the appendix tables to these names.

Machines. Every machine is a rented host with one RTX 5090. Each runs our bandwidth probe, which measures how fast it reads host memory from CPU threads (B_c), over PCIe (B_p), and by any method (B_{host} , the highest single reading). The *link-to-CPU ratio* is B_p/B_c . Ryzen, Core and Threadripper parts are *consumer* processors; EPYC, Xeon and engineering-sample parts are *server* processors.

Statistics and registration. Time per token is averaged over problems, and speed ratios are ratios of mean times. Machines differ far more than problems: two panel machines with the same CPU model differ by up to 29% in the deployed cache’s time, while a machine rented again came within 3.2% of its earlier run. The machine is therefore the unit. Claims are about orderings that hold on every machine of a stated class, and summaries over machines are bootstrapped over machines (Kalibera & Jones, 2013). A *launch* is one rental; in the later experiments each configuration also ran in two or three separate processes per launch. A machine *ran stably* when its outputs matched the reference and, where it ran more than one process, their times agreed within 2%. Every experiment’s predictions were committed to a public branch before its machine started; Appendix C scores all of them.

Table 2. Cache configurations. *Reads*: how many times an admitted expert is read from host memory. All run in the same engine; the oracles read the future routing from a trace of the same text recorded on the same machine.

Name	What it admits, and how	Reads
<i>No foresight</i>		
deployed	decayed request count beats the lowest resident’s by $\kappa=1$; the CPU serves the miss, then a background copy fills the slot	2
single read	the same admissions, fetched into the slot in the step	1
admit every miss	every miss fetched into the lowest-scored slot (demand fetch)	1
<i>Oracles</i>		
MIN, 2 reads	MIN with bypass (greedy rule); the CPU serves the miss, then a copy	2
MIN, 1 read	MIN with bypass (greedy rule); admissions fetched in the step	1
MIN, fewest adm.	the hit-optimal schedule with the fewest admissions (an LP per host), with 2 or 1 reads	2 or 1
MIN, prefetched	MIN, 1 read, plus copies issued three steps ahead	1
no bypass	every expert of the coming steps, into the slot needed furthest ahead: Be-lady’s rule without bypass (also with 1 read, and prefetched)	2 or 1
window W	admit every miss unless the next W tokens show the victim needed sooner; victim the lowest-scored resident not seen in the window, else the one seen furthest	1
window W , re-call r	the same, each future expert kept with probability r , else replaced by a random one	1
deployed + win-dow W	the deployed policy with the same window and victim choice; the CPU serves the miss, then a background copy	2

Yardsticks. Four recur, one per section. Section 3 gives a system’s speed as a share of a bound’s speed. Section 4 splits the *gap*, the deployed cache’s time beyond Eq. (1), into shares. Section 5 gives speeds relative to the deployed cache on the same machine. Section 6 gives a window’s share of what full foresight saves over admitting every miss.

3 HOW FAST COULD IT BE?

Consider any system that executes the model’s exact routing with at most C experts of each layer resident on the GPU. Per token, each of the Lk routed experts is served by a resident copy or read from host memory. Let R^* be MIN’s host reads per token, summed over layers; no policy, even one that knows the future, reads fewer. If c experts per token run on the CPU, host memory supplies at least $\max(R^*, c)S$ bytes for experts of S bytes, and the GPU reads the dense

weights D plus $(Lk - c)S$. With everything overlapped,

$$\bar{T} \geq \min_{0 \leq c \leq Lk} \max \left(\frac{D + (Lk - c)S}{B_{\text{gpu}}}, \frac{\max(R^*, c)S}{B_{\text{host}}} \right). \quad (1)$$

At a host-bound budget this is MIN’s read time, R^*S/B_{host} . The bound assumes exact routing, whole experts, at most C residents per layer and one token per step. It drops all latencies, lets the two read paths share B_{host} , and is relative to the probe: a reader faster than the probe could beat it (Section 9).

A tighter bound when reads wait for the router. A system that reads an expert only after its layer’s router has chosen it cannot overlap that read with the GPU’s work: layer l ’s misses are known only after its attention and router, and layer $l+1$ starts only after layer l ’s experts. Its time is at least the GPU’s non-expert work plus the reads, in series:

$$\bar{T}_{\text{demand}} \geq T_{\text{GPU}} + \frac{R^*S}{B_{\text{host}}}, \quad (2)$$

where T_{GPU} is the GPU’s attention, dense layers, norms and routers per token: 2.9 ms for gpt-oss, the smallest of our engine’s Nsight profiles. Every system we compare reads on demand; only reads issued before routing can go below Eq. (2).

A bound that respects the layers’ order. Foresight does not free every read. An expert run on the CPU needs its layer’s attention output, and the next layer’s attention needs its result, so CPU reads sit in series with the GPU’s non-expert work; only copies over the link can be issued ahead. With a share f of MIN’s reads served by the CPU at its best rate B_c and the rest copied at the link’s best rate B_p ,

$$\bar{T} \geq \min_{0 \leq f \leq 1} \max \left(\frac{(1-f)R^*S}{B_p}, T_{\text{GPU}} + \frac{fR^*S}{B_c} \right), \quad (3)$$

and never below Eq. (1). On the median consumer machine at gpt-oss 11%, Eq. (3) equals Eq. (1) (ratio 1.00); where the link is slow, it is up to 1.56 times as high.

The read time is nearly reachable (registered). A microbenchmark replays MIN’s per-layer reads of gpt-oss on 3 machines without the model: CPU threads read part of each layer’s experts while the copy engine copies the rest. It reaches 86–96% of the bound’s read time (Table 29). The GPU term of Eq. (1) is looser, but it binds only at the largest budgets (Table 28).

Where systems stand. Table 4 places three systems against the bounds on two machines. Our cache leads FreeToken (Yang et al., 2026) at 11 of 12 configurations and runs 2.0–4.0 $\times$ as fast as stock llama.cpp with the same GPU

Table 3. The deployed cache’s gap to Eq. (1), split by oracles in the engine (Fig. 2): mean share of the gap over the 13 machines whose link reads at least 0.5 of their CPU rate (a split drawn after the data), with 95% intervals over machines. The first two rows each change one thing; the third changes both, and *together, ahead*, T_{GPU} and the residual sum to the gap. T_{GPU} is the median Nsight profile, not removed in the engine; with the smallest and largest profiles its share at 11% is 27–31%.

Part of the gap	gpt-oss 11%	25%
MIN’s set alone, still read twice	1% [−1, 3]	21% [19, 22]
One read alone, deployed set	1% [0, 2]	−1% [−4, 1]
Both together: MIN’s set read once	39% [32, 45]	35% [30, 40]
Then reading ahead: copies issued a few steps early	15% [11, 18]	20% [18, 22]
GPU’s non-expert work in series, T_{GPU} (assumed)	29% [26, 32]	36% [33, 39]
Residual above Eq. (2) with full foresight	18% [12, 23]	8% [4, 13]

memory for experts. Tuned on a third machine, FreeToken still trails at 5 of six cells (Appendix M). Across 25 machines with consumer processors at gpt-oss 11%, our cache runs at 31–54% of the speed of Eq. (1), 40–54% of Eq. (3) and 55–70% of Eq. (2). These shares are relative to one probe run: its second-highest reading is a median 0.8% below the highest, and on the machine at the top of the range the engine’s reads imply a rate above the probe’s best (Table 27).

The field leaves most of the machine unused. To compare with published systems, we score each one against the same bound computed from its paper’s own hardware at datasheet rates. Over 20 published batch-1 measurements, the median is 13.6% of the bound; our cache, scored the same way, reaches a median 27% (Appendix N). Where we lack a model’s routing trace, uniform routing stands in, which overstates the bound’s reads and so the published percentages.

4 WHERE THE SECONDS GO

The gap is the deployed cache’s time beyond Eq. (1). Oracles in the engine remove three parts of it: what is cached (MIN’s set instead of the deployed one), how it is read (each admitted expert once instead of twice) and when (copies issued a few steps ahead). We change the first two alone and together. The GPU’s non-expert work in series with the reads is not removed in the engine; we take it from the profiles (T_{GPU} , median 3.2 ms), and the rest is a residual. Figure 2 shows the 15 machines that ran every state, and Table 3 the mean shares on the 13 with fast links, a split we drew after the data.

Table 4. Decode speed at equal GPU expert memory on two RTX 5090 hosts (30 AIME-25 problems, the first 256 greedy tokens, one session). Ratios are of mean speeds, paired by problem, with 95% bootstrap intervals over problems within one launch on one machine; each system decodes its own greedy text. Our cache uses each machine’s fetch table; FreeToken uses its faster backend per budget, picked on host B on a separate launch. *Bound*: Eq. (1) on that machine (MIN’s reads, the host’s highest probed rate) with the GPU at its datasheet rate, and at the rate all-in-VRAM decode measures at batch size 1 (52% of datasheet for gpt-oss, 61% for Qwen3; Table 28 varies the other assumptions). Against the pooled bound their designs allow, FreeToken (one cache for all layers) and llama.cpp (whole layers pinned) stand at 18–40% and 8–20%. With every weight in GPU memory (an RTX PRO 6000) stock llama.cpp decodes gpt-oss at 255 and Qwen3 at 178 tok/s. [†] Selection launch only, which favours FreeToken.

Model	Experts on GPU	llama.cpp (tok/s)	FreeToken (tok/s)	Ours (tok/s)	Ours ÷ FreeToken (95% CI)	Ours ÷ llama.cpp	Bound, datasheet tok/s	GPU ours	Bound, measured tok/s	GPU ours
Host B: Ryzen 9 9950X3D, card at 17001 MHz memory clock, CPU 72 GB/s, link 53, together 78, highest (the bound’s rate) 88										
gpt-oss-120b	11%	34.9	54.0 (hybrid)	69.9	1.294 [1.278, 1.312]	2.00×	172	41%	172	41%
gpt-oss-120b	25%	40.0	85.6 (hybrid)	109.2	1.275 [1.254, 1.295]	2.73×	430	25%	279	39%
gpt-oss-120b	40%	47.7	132.2 (offload)	152.5	1.154 [1.134, 1.172]	3.20×	519	29%	279	55%
Qwen3-30B-A3B	12.5%	19.5	38.7 (hybrid)	40.0	1.032 [1.022, 1.042]	2.05×	96	42%	96	42%
Qwen3-30B-A3B	25%	22.1	54.7 (hybrid)	63.1	1.153 [1.134, 1.173] [†]	2.86×	214	30%	192	33%
Qwen3-30B-A3B	43.75%	28.4	103.1 (offload)	108.2	1.049 [1.036, 1.064]	3.81×	308	35%	192	56%
Host S: Ryzen 9 9950X at the stock clock, card at 14001 MHz, CPU 56 GB/s, link 53, together 66, highest 71; FreeToken’s backend carried over from host B										
gpt-oss-120b	11%	28.9	48.0 (hybrid)	57.9	1.207 [1.195, 1.219]	2.00×	140	41%	140	41%
gpt-oss-120b	25%	33.6	78.7 (hybrid)	94.1	1.196 [1.163, 1.224]	2.80×	351	27%	275	34%
gpt-oss-120b	40%	40.1	122.2 (offload)	133.6	1.093 [1.077, 1.111]	3.34×	514	26%	275	49%
Qwen3-30B-A3B	12.5%	16.1	32.7 (hybrid)	33.5	1.027 [1.018, 1.036]	2.08×	78	43%	78	43%
Qwen3-30B-A3B	25%	18.5	51.4 (hybrid)	53.9	1.048 [1.037, 1.059]	2.91×	174	31%	174	31%
Qwen3-30B-A3B	43.75%	23.8	98.3 (offload)	95.8	0.974 [0.962, 0.987]	4.02×	305	31%	189	51%

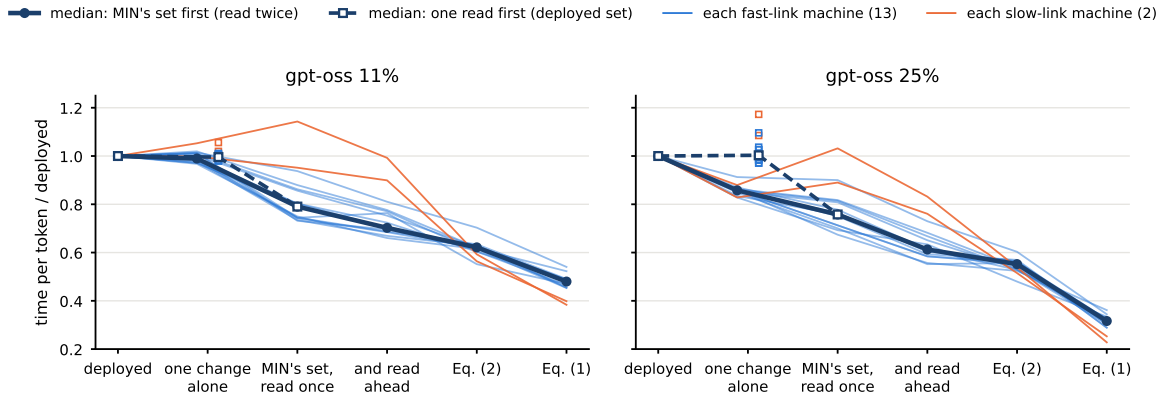


Figure 2. Where the seconds go, measured: the time per token of each state, relative to the deployed cache on the same machine, as the oracles remove one part of the gap after another, down to the two bounds. One line per machine that ran every state, one launch per GPU. On the two machines with slow links (orange), reading once in the step saves little or costs time.

What is cached and how it is read pay only together. At gpt-oss 11%, caching MIN’s set while still reading each admission twice closes 1% of the gap, and reading each admission once while caching the deployed set closes 1%. Doing both closes 39% (Fig. 2). MIN’s set admits more experts than the deployed one, and each extra admission costs a second read unless it is read once. At 25%, where admissions are fewer, MIN’s set alone closes 21% and reading once brings it to 35%.

Reading ahead. Issuing MIN’s copies a few steps ahead closes another 15% of the gap at 11% and 20% at 25%. With all three parts removed, the engine’s time on these machines is 1.07–1.38 times Eq. (2) at 11%.

The GPU’s own work in series. The GPU’s non-expert work, in series with the reads, is 29% of the gap at 11% (27–31% over the profiles’ range) and 36% at 25%. No oracle we built removes it: the misses that remain are still read after their router runs. Removing it needs every read issued before routing, which only copies over the link allow (Eq. (3)). The residual is 18% of the gap at 11% and 8% at 25%.

Slow links change the picture. On the two machines whose link reads at most 0.40 of their CPU rate, reading once in the step saves little or costs time, and the best oracle closes at most 17% of the gap at 11%. Where the link is fast, the best oracle closes a mean 54% (95% interval over

machines 49–58). Section 5 shows what does pay on slow links.

A closed-form account (exploratory, not confirmed).

On consumer processors, the deployed cache’s time is close to the GPU’s profiled kernel time plus its counted host reads at B_{host} : at gpt-oss 11% the read rate its time implies is $0.88\text{--}1.16$ of B_{host} over 39 launches. We registered it, first in a plainer form, before four experiments: three failed and one was inconclusive by its own rule. It held on machines we had rented before and missed by up to 11% on a new consumer machine (Appendix K). The measured decomposition above does not rely on it.

5 HOW FORESIGHT MUST BE SPENT

This section measures each way of spending foresight as a speed relative to the deployed cache on the same machine. A running example: on a Ryzen 9 9950X at gpt-oss 11%, Eq. (1) gives 10.0 ms per token and Eq. (2) 13.0. The deployed cache takes 20.6, MIN read once in the step 15.2, and MIN read once and ahead 13.8.

Read each admitted expert once (registered). MIN read once in the step is faster than the deployed cache at every budget on the three factorial machines, by 16–51%, and by up to 81% when read ahead (Fig. 3). The usual ways of spending foresight waste reads. Prefetching without bypass, Belady’s original rule, admits experts MIN would bypass. Serving a miss on the CPU before copying it reads every admission twice. At the smallest budget of each model these gain at most 15% or lose, a trade-off known for disks (Cao et al., 1995) and CPU caches (Jain & Lin, 2018).

Admit as rarely as optimality allows (registered). The greedy schedule admits whenever MIN allows and may evict an expert again before its next use. The fewest-admission schedule makes 0.60–0.73 of the greedy one’s copies for the same hits in replay (in the engine it misses 2.5–5.3% more). Read once in the step at gpt-oss 11%, it beat the deployed cache on all 9 machines that ran it stably, by a geometric mean of $1.21\times$ (95% interval over machines 1.13–1.30). The gain was registered on 7 of them and observed on the others. Counting the four machines that ran unsteadily as well, the mean is $1.17\times$ over 13; it lost on two of them, the two with the slowest links.

The machine decides the read path (exploratory).

Across machines, the gain of every configuration that moves reads from the CPU to the link follows the link-to-CPU ratio: for MIN read once in the step, Spearman 0.89 [0.63, 0.97] over 19 machines (Fig. 4). The machine carries 96% of the variance of that gain; problems carry at most 3%. Where the link reads at 0.28–0.32 of the CPU rate, the greedy schedule

read once in the step lost at 11% ($0.85\text{--}0.93\times$). There the fewest-admission set pays when the CPU serves it and the copies run in the background: $1.11\text{--}1.22\times$ over the two budgets.

Without foresight: admit less (registered, partly held).

The fewest-admission schedule wins because an admission costs a read. The deployed cache’s admissions are second reads, so the lesson applies to it without foresight. Raising its admission margin to the value that minimises its reads on other text cuts its host reads by 4–6% at gpt-oss 11%. On the 5 machines that ran it stably, it runs $1.021\times$ the deployed cache at 11% and $1.018\times$ at 25% (geometric means over machines). The gain is small because admissions are a small share of the deployed cache’s reads (Appendix L).

6 HOW MUCH FORESIGHT IS NEEDED

The oracles know the whole future; a predictor would know part of it, imperfectly. We give the cache a *window*, the routing of the next W tokens, exact or with each future expert right only with probability r . Throughout this section, the value of a window is its share of MIN’s gain over admitting every miss, the online policy that reads least at these budgets (Appendix J gives the policies).

Next-token prediction buys little. Exact routing of the next token recovers 0.18 of MIN’s gain at gpt-oss 11% and 0.07 at 25%, and at most 0.31 at any host-bound budget of our two models (Fig. 5). Half of the gain needs 4 and 10 tokens. In the engine, an exact 16-token window recovers 0.78–0.93 of the gain in time at 11% (registered at 0.80 or more, which the lowest machine missed).

Accuracy matters as much as horizon. An 8-token window that is right half the time recovers 0.29 of the gain at 11%, about what 2 exact tokens buy (0.31).

The horizon is measured in distinct experts (exploratory).

On 9 models’ own sampled text, the window that closes half of the read gap ranges from 0.6 to 34 tokens. Paging theory measures lookahead in distinct pages (Albers, 1997; Breslauer, 1995), and so does the rule that fits: within that window a layer requests about $0.65 C$ distinct experts (quartiles 0.61–0.72). On the AIME routing the engine runs, the same quantity is 0.66–0.81 C . Given a model’s routing trace, this one-constant rule predicts its horizon as well as a two-constant power law in tokens (Appendix I).

Nothing realisable reaches it. Forecasters trained on routing histories, linear or recurrent, close at most 15% of the read gap on gpt-oss: their precision falls with the horizon. Speculative batches save an online cache 8–12% of its reads at 90% draft acceptance and cost 18–20% at 80% (Chen

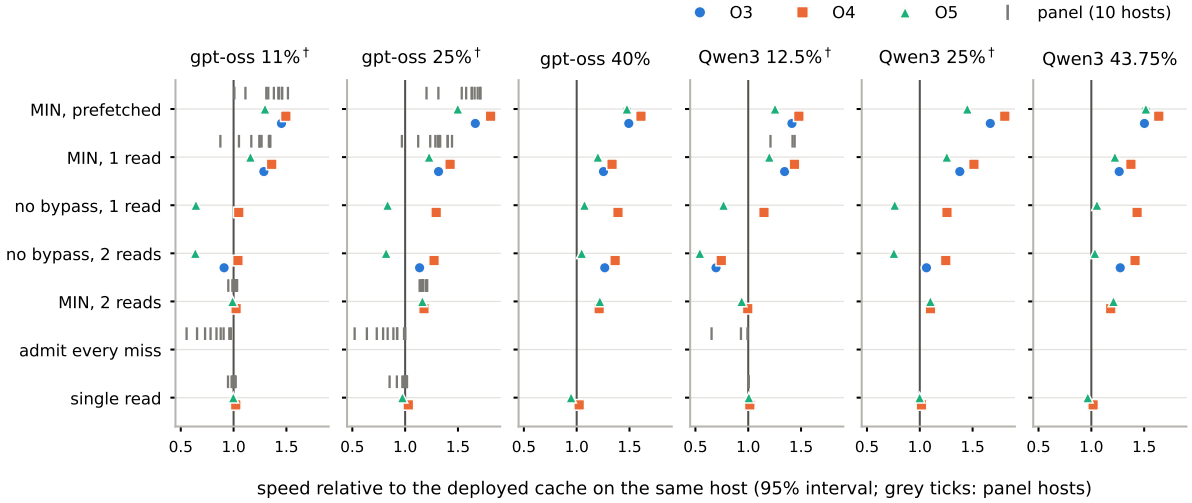


Figure 3. What foresight is worth, by how it is spent: speed of each configuration of Table 2 relative to the deployed cache on the same machine, at the six budgets ([†]host-bound). Three machines at every budget (O3–O5: two Ryzen 9 9950X and a 9950X3D); the 10 panel machines (grey ticks) at the two host-bound gpt-oss budgets, and 3 of them at Qwen3 12.5%. *MIN, 1 read* gains at every budget on these machines; reading twice or admitting what *MIN* bypasses gains at some budgets and loses at others.

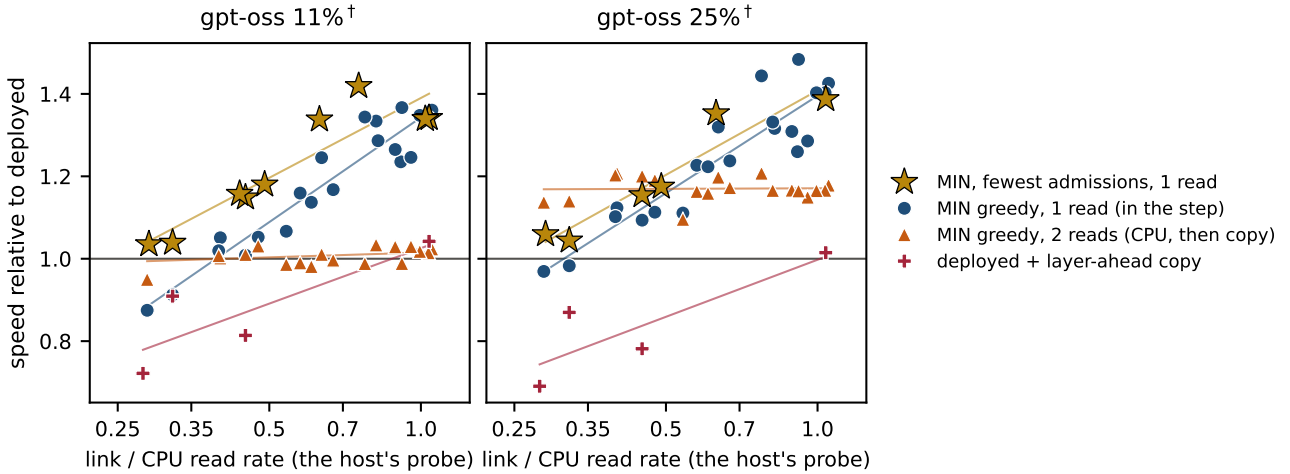


Figure 4. The machine decides which way of spending foresight pays: speed relative to the deployed cache on the same machine against the machine’s link-to-CPU ratio, at the two host-bound gpt-oss budgets. Every marker is a machine that passed our checks. Stars: *MIN*’s fewest-admission schedule read once in the step. Circles and triangles: *MIN*’s greedy schedule read once (in the step) or twice (by the CPU, then copied). Crosses: the deployed cache with the engine’s layer-ahead copy. Lines: least-squares fits in the log of the ratio, for the trend only.

et al., 2025b; Li et al., 2026; Saxena et al., 2026). Layer-ahead predictors (Hwang et al., 2024; Zhu et al., 2025) reach 84% recall in our engine but see less than one token ahead. The engine’s layer-ahead copy gains $1.04\times$ where the link is as fast as the CPU and loses elsewhere, down to $0.72\times$ at 11% (registered; Table 23).

7 WHAT THIS MEANS FOR SYSTEM BUILDERS

1. **Report the distance to the machine’s bound.** A speed-up over a chosen baseline hides how much is left. Equation (1) needs only the routing trace and one probe run, and on our machines the best system sits at a third to a half of it at the smallest gpt-oss budget.

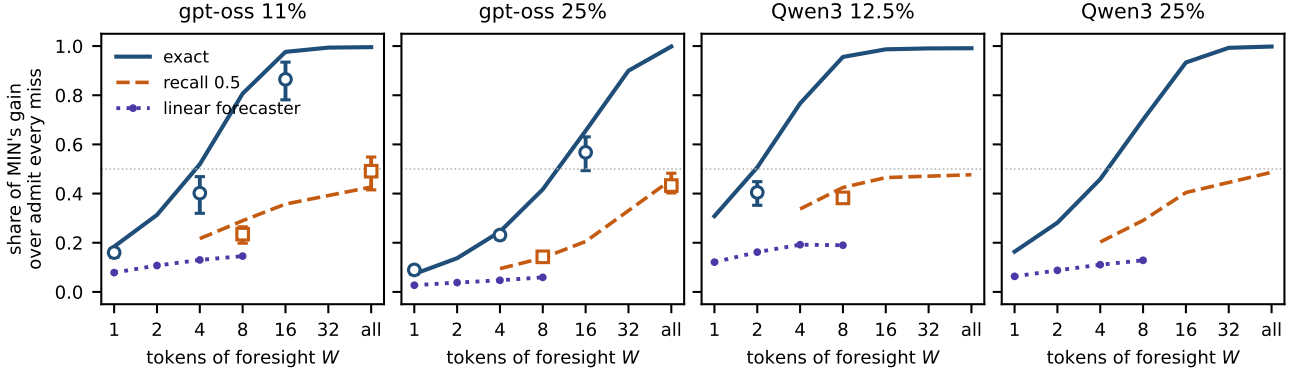


Figure 5. The value of partial foresight at the four host-bound budgets: the share of MIN’s gain over *admit every miss* recovered by exact routing of the next W tokens (solid), by a window whose experts are each right with probability 0.5 (dashed), and by a linear forecaster fitted on other text (dotted), in host reads on the routing the engine runs. Markers: the same share in time per token on the panel machines (mean and range over machines; 10 machines for gpt-oss, 3 for Qwen3 at 12.5%). The grey line marks half of the gain.

2. Change what is cached and how it is read together.

Serving a miss on the CPU and then copying it into a slot reads it twice. At the smallest gpt-oss budget a better set read twice, or the same set read once, closes almost none of the gap; a better set read once closes 39%.

3. Admit less.

Every admission is a read. Among equally good schedules, the one with the fewest admissions gains where links are slow and ties where they are fast; online, a higher admission margin gains $1.021\times$ on average at 11%.

4. Choose the read path by the link-to-CPU ratio.

On fast links, copy misses into slots in the step. On the machines we ran at a ratio of 0.32 or less, serving them on the CPU and copying in the background paid instead; few such machines ran stably, so where the crossover lies is uncertain. The probe that measures the ratio takes seconds.

5. Issue reads before routing, and predict far enough ahead to save reads.

A quarter to a third of the gap is the GPU waiting in series with reads that only a prediction of the routing lets a system issue earlier, and only over the link (Eq. (3)). To save reads, predictors should target the next $\approx 0.65 C$ distinct experts of each layer ($0.66\text{--}0.81 C$ on our workload), not the next token; accuracy counts as much as horizon.

8 RELATED WORK

Offloading systems. Demand-paging systems cache experts per layer and copy misses over PCIe (Eliseev & Mazur, 2023; Xue et al., 2024; Song et al., 2024a; Tang et al., 2024; Kong et al., 2024; Zhong et al., 2024; Yi et al., 2023; He

et al., 2024; Shams, 2026), or use predicted expert activations to decide what to keep (Du et al., 2024); an upstream llama.cpp pull request adds such a cache, admitting every miss into an LRU (Gupta, 2026). CPU-execution systems run non-resident experts on the host (Kamahori et al., 2024; Chen et al., 2025a; Zhong et al., 2025; Zhang et al., 2025; Wang et al., 2026a; ATSIInfer), as PowerInfer runs a dense model’s cold neurons (Song et al., 2024b); a set-associative GPU cache with CPU-executed misses (Huang et al., 2025), a llama.cpp fork (leloch, 2026a;b) and two llama.cpp pull requests (miltos22, 2026; csantiago78, 2026) are the closest designs to ours. FreeToken (Yang et al., 2026) and DALI (Zhu et al., 2026) split each layer’s misses between CPU and PCIe. Predictors guess upcoming experts from the previous layer’s state (Hwang et al., 2024; Zhu et al., 2025), several tokens ahead with a learned forecaster (Wang et al., 2026b; Yadav & Asgari, 2026), or from draft tokens (Chen et al., 2025b; Li et al., 2026; Saxena et al., 2026); Read-ME decouples the router so routing is known in advance (Cai et al., 2024). Batched-throughput systems (Sheng et al., 2023; Cao et al., 2024; Yuan et al., 2025) and flash or SSD tiers (Alizadeh et al., 2024; Liang et al., 2026a; Lin et al., 2026) are outside our scope.

Bounds, oracles and caching theory. WiSP (Zhang et al., 2026b) bounds decode by PCIe turnover, Budgeting Bytes (Zhang, 2026a) by bytes over bandwidth for a given fetch set, and MoE-CAP (Jiang et al., 2025) measures bandwidth utilisation against a single peak; the roofline (Williams et al., 2009) is the single-memory ancestor of our bound. Zhang (2026b) propose a trace-level evaluation contract and report a 44–46% gap between causal policies and MIN with bypass; Liang et al. (2026b) measure, across 20 models, the hit rate of an oracle cache that sees m tokens ahead. We add

the time domain: oracles inside a real engine, a measured decomposition of the gap, the cost of how foresight is spent, and a horizon rule in distinct experts. SAEM (Zhang et al., 2026a) gives its own stage-level planner perfect knowledge of the coming stage and finds that worth at most about 14% on an A100; SpecMD (Hoang et al., 2026) benchmarks expert-cache and speculative prefetch policies across hardware; Angelopoulos et al. (2025) give competitive bounds for layered expert paging. With bypass, MIN maximises hits exactly (Carlisle & Lloyd, 1995; Jain & Lin, 2016); when bypass and admission cost differ, the time-optimal placement is a different problem (Zhang et al., 2020), which is why we use MIN’s reads as an input to a bound rather than as a time-optimal policy. Choosing among hit-optimal schedules by what admissions cost has precedents: Demand-MIN for prefetching (Jain & Lin, 2018), offline-optimal caching as a min-cost flow over reuse intervals (Berger et al., 2018), and admission rationing in flash caches (Eisenman et al., 2019). Our fewest-admission schedule is that interval formulation with admissions as a second objective, measured in an engine. Learning-augmented paging is most robust with predictions of whether a page will be evicted before its next use (Antoniadis et al., 2023); learned replacement imitates Belady in CPU and content-delivery caches (Jain & Lin, 2016; Song et al., 2020) and in an SSD-backed expert cache (Kim et al., 2026).

9 LIMITATIONS

- **Hardware and sample.** One GPU model, rented, mostly in desktop hosts; clocks could not be locked and power limits differed. The machines are a convenience sample of the cheapest offers that passed our checks, and few have a link slower than half their CPU rate, so where the crossovers lie rests on few points. Rented machines are often shared or unsteady: in the two tests on new machines, 3 and 2 machines stopped at the gate and 2 and 1 ran unsteadily, and of 8 launches on server processors only three passed every check.
- **Workload.** Batch-1 decode of two models on AIME prompts, also used while developing the system, teacher-forced on each model’s own greedy text. Prompts and batches above one bypass the cache.
- **The bounds.** Exact routing and per-layer slots; pooling slots across layers lowers MIN’s reads by 4.5–18.6%. R^* is counted on the 30-problem trace; on the first 20 problems, which the later experiments time, it changes by -0.4% . The bounds drop latencies and are relative to one run of our probe, which on most machines ran while the model was downloading in the background: by the closed-form account of Appendix K, the engine’s reads imply a rate more than 5% above the probe’s highest reading on 12 of 84 launch-budgets

(Table 27). T_{GPU} is our engine’s kernel time, not a property of the machine alone.

- **The decomposition.** The parts are measured at two budgets of one model, and its summary is over machines with fast links, a subset we drew after the data. T_{GPU} is profiled rather than removed in the engine, and the residual absorbs any error in it. The machines pool runs of 20 and 30 problems and three versions of the engine. Equation (3) assumes that the CPU reads an expert’s weights when it runs the expert.
- **The oracles.** Oracles read a recorded trace. In the engine the fewest-admission schedule misses 2.5–5.3% more than the greedy one, more than its replay predicts; its measured gains include that cost. Degraded windows draw errors independently, while a real predictor’s errors correlate with what the cache holds.
- **Outputs.** Parity is measured by teacher-forced KL divergence against an all-VRAM reference (mean 0.0019 and 0.0005 nats), not by task accuracy.

10 CONCLUSION

MIN’s host reads at a machine’s probed rate bound batch-1 MoE decode with experts in host memory, and the bound’s read time is nearly reachable. Our cache runs at 31–54% of it at the smallest gpt-oss budget, and the other systems we measured run further from it. Oracles inside the engine show where the rest goes: at the smallest budget, caching MIN’s set and reading each admitted expert once pay only together, reading ahead adds to them, and the GPU’s own work waiting in series with the reads is most of what remains. Where the link is fast, foresight spent as MIN spends it, each admitted expert read once and ahead, removes about half of the gap; where the link is slow, admissions must be read by the CPU and copied in the background, and foresight removes less. Its value has a price: on our workload half of it needs the routing of 0.66–0.81 C distinct experts per layer ahead, well beyond the next token.

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A SCIENTIFIC BENCHMARKING CHECKLIST

How this paper follows the twelve rules of [Hoeffler & Belli \(2015\)](#).

Rule	Where and how
1. Report the base case of a speed-up and its absolute performance.	Every ratio in Table 4 is next to both absolute speeds; llama.cpp is configured with per-layer expert offload at equal expert memory, the setting its users run.
2. Explain subsets of benchmarks or resources.	Two models chosen for size and precision (MXFP4, BF16), a third that is FreeToken’s own headline model; budgets span what fits a 32 GB card; the 43.75% budget is omitted where it does not fit. All 30 AIME-25 problems are used.
3. Arithmetic mean for costs, harmonic mean for rates.	The accounting, the panel and the value of foresight average time per token over problems (a cost) and quote speeds as the harmonic mean this implies; Table 4 averages per-problem rates arithmetically, which departs from this rule; every problem decodes the same number of tokens, so the harmonic mean is the right summary, and ratios of total time agree with the reported ratios of mean rates within 0.003.
4. Avoid summarising ratios.	Ratios of means are never averaged across budgets or models; ranges list the per-budget values, and the medians reported across models (Table 16) or across published rows (Appendix N) are medians of per-cell quantities, labelled as such.
5. Report whether measurements are deterministic; give confidence intervals.	Decode is greedy and deterministic per engine; timing is not. Every comparison has a 95% interval over problems; the panel (job 099) adds intervals over hosts, the unit that varies most.
6. Do not assume normality.	Intervals are bootstrap percentile intervals (10,000 resamples, paired by problem); no distributional assumption.
7. Compare nondeterministic data soundly.	Paired bootstrap of the ratio of means within a host; across the panel’s hosts, a two-stage bootstrap (hosts, then problems) (Kalibera & Jones, 2013); the variant a system is scored with is picked on a separate launch (except the Qwen3 25% row of Table 4).
8. Check that central tendency is the right summary.	Mean speed over 256 tokens is FreeToken’s own metric; per-token p50 and p99 latencies are recorded for every problem in the released rows.
9. Document all factors and the setup.	Machines, probes, power limits, memory clocks (Section 2), software commits (llama.cpp 4da6337, FreeToken 0d652e7), flags and job scripts are released; Appendix O.
10. Report measurement, synchronisation and summarisation.	One client times every system of Table 4 from streamed token arrivals; the oracle comparisons of Section 5 are timed in process by llama-ec-bench over teacher-forced text, every policy by the same timer (Section 2).
11. Show upper performance bounds.	The bound (Section 3) is a column of Table 4; Table 28 tightens it four ways; Section 3 applies it to the published literature and to our cache with the same procedure.
12. Plot enough to interpret; connect points only for trends.	Figure 3 plots every host’s ratio with its interval; Fig. 5 connects measured windows for the trend only and marks engine measurements separately; Figs. 8 and 11 are bar charts and Figs. 6, 7 and 12 scatter plots.

B NAMES AND TERMS

Table 6 maps the configuration codes used in the appendix tables, the supplement and the scripts to the names of the main text; Table 7 defines the other terms.

Table 6. The configuration codes of the appendix tables, the supplement and the scripts, and their names in the main text (Table 2).

Code	Main-text name, and what it is
base	the deployed cache
foa	single read: the deployed admissions, fetched into the slot in the step
aa	admit every miss
dk	the deployed cache with a larger admission margin (admitting less)
lrn	the deployed cache ordered by a learned reuse predictor
pf	the deployed cache with the engine’s layer-ahead copy
fetch	MIN, 1 read: the greedy schedule, admissions fetched in the step
bypass	MIN, 2 reads: the greedy schedule, served by the CPU, then copied
fetchplan	MIN’s fewest-admission schedule, 1 read
bypassplan	MIN’s fewest-admission schedule, 2 reads
lead2	MIN, 1 read, copied two steps ahead of use
both2, both3p	MIN, prefetched: fetch plus copies two (three, paced) steps ahead
hitopt, hitoptp	no bypass, 2 reads (paced)
nb2	no bypass, 1 read, copied two steps ahead
w4, w16	window $W=4, 16$: admit every miss unless the window shows the victim needed sooner
w1, w2	admit every miss, with exact foresight of the next 1 (2) steps
w8r5, allr5	admit every miss, with foresight of 8 steps (the rest of the problem), each future expert seen with probability 0.5
b4, b16	deployed + window $W=4, 16$
b8r5	deployed + window $W=8$, each future expert seen with probability 0.5
learned	single read with the learned admission and victim order (job 097; not the same as lrn)

C PREDICTION LOG

Each job’s predictions are in the header of its script on the public `gpu` branch, committed before the machine started; each machine cloned the published branch at boot and ran the header as committed, so git history orders every prediction before its measurement (for seven of the 71 rentals of jobs 093–108 the push and the rental’s creation fall within a second of each other, and for four GitHub records the push up to 0.8 s after the rental was created; every commit preceded its rental by at least 2.7 s, and every machine started its job at least 26 s after the push; `scripts/reg-timing.py`). Table 8 summarises every job’s clauses; the notes below give the failures that matter. Outcome notes with the statistics are in `prereg/`. The scorecard, generated from `prereg/scorecard.clauses.json`, splits every prediction of jobs 073–098 into its separable clauses and scores each under one rule (jobs before 073, which built and debugged the engine, also registered predictions in their headers; their outcome notes are in `prereg/`, and they are not scored here; job 072’s, for one, failed or did not apply on two of its three): *held* only when the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it); *held (point)* when only the point estimate is, or no interval exists (except in the hand scoring of jobs 073–098, which holds deterministic counts such as reads outright, since they have no sampling error); *failed* when the point estimate is on the wrong side; *untested* when the clause could not be evaluated; *void* when the job was voided. Of 565 clauses, 218 held, 193 held on the point estimate, 142 failed, 8 were untested and 4 void: 7 of 19 scored clauses held in jobs 073–075, when the system was being built, 19 of 23 in jobs 076–081, 11 of them sign-only predictions made after a same-host pilot, 24 of 38 in jobs 082–087, and 168 of 473 in jobs 088 onward, where 42 of 82 sign clauses held (75 on the point estimate) and 90 of 301 bands failed.

Failures, jobs 073–098. Most bands that failed are the sizes of effects whose direction held: our slow-link lead landed at $1.51\text{--}1.58\times$ against a predicted $1.10\text{--}1.50$; the rerun of Table 4 on a second host (job 089) gave ratios $0.06\text{--}0.11$ below host B’s at five of six cells against a predicted ± 0.06 , host-bound speeds $14\text{--}17\%$ lower against a predicted $\pm 12\%$, and LRU $6\text{--}8\%$ slower than decayed frequency at the two GPU-heavy cells against a predicted $10\text{--}30\%$; on the 4090 and 3090 hosts (jobs 091, 092; Appendix E) our lead over FreeToken and its fraction of the limit overshot their bands, FreeToken’s carried-over backend fell far below its band, and two launch-order checks exceeded 3% (7% and, at gpt-oss 11% on the 3090 host, 11%); in the first oracle job (093; Section 5) the oracle’s gain fell outside $20\text{--}80\%$ at the two lowest budgets (it lost), its speed fell just below the predicted $45\text{--}65\%$ of the limit at all four host-bound cells ($43.7\text{--}44.8\%$), its hit rate and admissions missed the optimum’s by more than the predicted 3 points and 30% at three and two of the four, no-overlap cost 0% against a predicted $5\text{--}25\%$, and all-CPU cost 15% on Qwen3 against a predicted $0\text{--}10\%$; in the bytes-oracle job (094) the bands missed as Appendix H describes; in the single-read job (095) 44 of 56 clauses held, and the 12 that failed are of size: the model on the fetch oracle’s counters was predicted within 10% and missed by $13\text{--}17\%$ at four cells, the fetch oracle’s reads

Table 7. Terms used throughout.

cell	a model at a budget: the share of each layer’s experts resident on the GPU
E, k, C, L	experts per layer, experts per token, GPU slots per layer, MoE layers
miss	a requested expert not resident: read from host memory by the CPU or by a copy
fetch	copy a miss into a slot and run it there in the same step (one read)
admission	making an expert resident; <i>serve-then-copy</i> : the CPU serves the miss, then a background copy reads it again
prefetch, lead d	copy an expert into a slot at least d steps before its first use (one read)
paced	copies in 16 MB pieces, one in flight
MIN, R^*	Belady’s optimum with bypass on the routing trace; its host reads per token, the fewest of any policy
online / oracle	a policy that sees only past routing / one that reads future routing from a recorded trace
$B_c, B_p, B_{cp}, B_{host}$	probed host read rates: CPU, link, both, highest by any method
bound, limit	Eq. (1): the speed bound of a machine for a cell (some appendix tables call it the limit)
host-/GPU-bound	the limit binds on host reads / on GPU work
$v(F), v(O, F)$	the modelled time with foresight alone / with foresight and overlap (Appendix G)
W_{50}	tokens of perfect foresight that close half of an online policy’s excess reads
hosts B, S	the two hosts of the system table (Table 4)
host A	the host of the ablation (Fig. 11)
O1–O6	the hosts of the oracle jobs 093–097 (Table 13, Appendix H)
Pa–Pj	the hosts of the panel, job 099 (Appendix C); later jobs return to Pe, Pf and Pg

were predicted within 10% of the simulation’s and came 17% below at the two GPU-bound cells (the registered simulator left a resident carried over from the previous sequence with no next use, and so evicted it first; corrected, the simulation is within 4% of the engine’s reads at every cell, Appendix H), the prefetching oracle was predicted at 55–85% of the limit at the host-bound cells and reached 44 and 51% at the two 25% cells, the model’s over-prediction of it was predicted at 15–50% and came to 7 and 13% at the two lowest budgets, and the paced variant, predicted within 10% of the unpaced one, was 12–13% faster at the two 25% cells; in the confirmatory factorial (096) the clauses held at most cells of both hosts and the failures are listed in Appendix H, the largest being the difference between the hosts; and in the FreeToken tuning job (098) its best setting stayed within the predicted 10% of its default and our ratios within the predicted bands, while the fixed fetch caps, the thread count and the CPU-only comparison missed as `prereg/freetoken.outcome.098.md` describes. Among the 23 threshold failures are ours at $1.56\text{--}1.78\times$ llama.cpp on the slow hosts against a predicted 1.8, the Mixtral tie against a predicted $1.3\times$, the hit-optimal oracle’s gain at the two GPU-bound cells (48 and 56%) against a predicted ceiling of 15%, the bytes-optimal oracle short of beating the hit-optimal one by 15% at the two lowest budgets, and its 16-token window short of keeping 70% of its gain at three cells, fetch’s gain differing between hosts O4 and O5 by more than 0.15 at five cells, Belady’s single-read prefetch above $1.05\times$ the online policy at Qwen3 12.5% on O4, and FreeToken’s CPU backend running close to stock llama.cpp’s where we predicted at least $2\times$ and $1.2\times$; the 3 equality failures are two of the 4090 host’s FETCH tables, which copy one more expert than predicted, and FreeToken’s best backend at gpt-oss 25% in job 098 (offload, not hybrid); the 7 sign failures are LRU behind FreeToken at gpt-oss 11%, where we predicted it would still lead, and the hit-optimal oracle slower than the online policy at gpt-oss 11% and Qwen3 12.5%.

Changes against the running log. Against the running log kept during the study, two clauses are downgraded (a llama.cpp clause of job 074b that the log counted as held was never run; job 085b’s own restatement of its fourth prediction, “at least 2 of 3”, failed at 1 of 3, though the pooled 4 of 6 held), job 073’s “identical outputs” is scored on fully identical outputs (0–3%) rather than on agreement of the first 160 characters (27–47%), and ten clauses the log counted as held, among them job 087’s first, hold on the point estimate only under the interval rule; `prereg/scorecard.outcome.md` lists every change. Job 085’s Qwen3 half ran as 085b on a second machine with a slow link after a hung conversion. Job 084 ran on a throttled card and was rerun, with its predictions unchanged, as 084b/084c. The batched-verification variants of Section 6 (rejected drafts free, routed like the true token, or routed at random) are in `prereg/batchk.outcome.md`.

Scoring checks. The clauses were scored by hand from the outcome notes. A script (`scripts/scorecard.auto.py`) re-scores every clause whose measured value is a number and whose threshold parses, 469 of 565: it agrees with the hand score on 432; of the 37 differences, 36 are the deterministic counts (reads, divergences, table entries) that the hand scoring holds outright and the script scores held (point), and 1 is a boundary value the hand scoring read unrounded (1.249 against a band starting at 1.25). Of the clauses the script scores held on the point estimate, 213 have no interval and 7 have one that

Table 8. Every registered job from 073 on and how its clauses scored: *held* with a 95% interval on the predicted side (or outright, for deterministic counts), held on the point estimate only, failed, and untested or void. Jobs 073–098 were scored by hand, the rest by script; the supplement lists every clause.

Job	What it tested	Held	Point	Failed	Untested
073	ours against FreeToken, same host	3	0	4	0
074	the 40% budget fixed	0	2	4	0
074b	server overhead	2	1	0	1
075	RTX PRO 6000 at 40–60%	2	0	1	0
076	the system table on an RTX 5090	2	1	0	0
077	competitors, gpt-oss	3	0	0	0
078	Qwen3, four systems	3	0	0	0
079	Qwen3 profile	1	3	0	0
080	the fetch split	4	0	0	0
081	the system table with the calibrated fetch split	6	0	0	0
082	held-out prompts and steady state (review run 1)	4	0	3	1
083	gpt-oss, long outputs on hard held-out problems	5	0	0	0
084	all-in-VRAM column and traces (voided: throttled card)	0	0	0	4
084b	all-in-VRAM column and traces, rerun	2	2	1	0
085	a slower PCIe link, gpt-oss	5	0	1	0
085b	a slower PCIe link, Qwen3	5	0	1	0
086	Qwen3.6	1	1	2	0
087	ablation with a static cache	2	1	2	0
088	slow link, fair comparison	5	3	3	0
089	the system table on a second host	9	7	12	0
090	output parity against all-in-VRAM (RTX PRO 6000)	10	0	0	0
091	the grid on an RTX 4090	11	3	5	0
092	the grid on an RTX 3090	6	1	7	1
093	first foresight oracles	9	2	21	4
094	oracles that minimise bytes	12	1	24	1
095	single-read oracles	44	0	12	0
096	the oracle factorial	29	97	12	0
097	the learned admission order	11	53	20	0
098	FreeToken tuned	22	15	7	0
099	the host panel	190	42	39	2
100	windows of foresight	41	122	27	26
101	the crossover by link ratio	6	19	0	0
102	the read-schedule microbenchmark	0	28	5	0
103	fewest-admission schedule (plan not run)	0	4	0	41
104	fewest-admission schedule	1	22	24	0
105	Eq. (6), plain form; layer-ahead copy; fewest-admission set	7	20	4	0
106	Eq. (6) with overlap; admitting less; slow links by round	21	112	48	0
107	Eq. (6) on new machines; admitting less; fewest-admission loads	11	22	13	0
108	Eq. (6) on new desktop-class machines	0	13	6	2
all		495	597	308	83

crosses the threshold.

Jobs 099–104. Job 099’s clauses are scored by machine from the start (`scripts/panel_099.py`): of 273, 190 held, 42 held on the point estimate, 39 failed and 2 were untested (`prereg/scorecard_099.json`); the two untested clauses are one host’s replay check, whose step failed when a Python package did not install. The failures concentrate on the two hosts whose link reads at less than half their CPU’s rate, where the bands, set from four earlier hosts none of which was link-starved, did not encode host dependence; and on the planning time of the noisy whole-future window (`prereg/panel_outcome_099.md`). Job 100 (`scripts/job100.py`; its header gained a ninth prediction, about the two replacement machines, after the first four machines had started; it compares the replacements with values one of which was already measured): of 216, 41 held, 122 held on the point estimate, 27 failed and 26 were untested. Most failures are one prediction, that the deployed-path windows miss within 10% as often as a replay that admits at once (their copies land late); the rest are the probe-only model on the half-width-link host and four band or ordering edges. The untested clauses are the planning time of the deployed-path windows, which the engine does not time, and the paired prediction for a host that never reached a timed step (`prereg/window_outcome_100.md`). Job 101 (`scripts/job101.py`; the lowest-ratio panel machine relaunched, and a second Ryzen 7 9800X3D behind a slower link): of 25, 6 held, 19 held on the point estimate, 0 failed and 0 were untested (`prereg/crossover_outcome_101.md`). Job 102 (`scripts/job102.py`; the read-schedule microbenchmark on three hosts): of 33, 0 held, 28 held on the point estimate (none has an interval), 5 failed and 0 were untested. Four failures are one prediction, that waiting per layer would cost at least 0.03 of the host term against the analytic schedule; on two of the three hosts it costs nothing measurable. The fifth is the ordering of the two budgets on the Intel host (`prereg/readsched_outcome_102.md`). Rentals that produced no results were destroyed and are in the rental ledger (`gpu/vast_ledger.json`): the first two for 099a, replaced before any measurement; 100d, which stalled downloading the model; 100e, which never reached its job; and 103c, which never started its job. Job 103 (`scripts/job103.py`; MIN’s fewest-admission schedule against the greedy one): the hosts could not install the plan’s solver beside Debian’s numpy, so the engine ran the greedy schedule in the plan configurations, as it logs; of 45 clauses 4 held on the point estimate (the relaunch of Pf) and 41 are untested. Job 104 repeated it with the plan in its own environment on three of the same machines (Pf, a second Core Ultra 9 285K whose link reads at 0.49 of its CPU rate, and Pg, whose second launch job 103’s 5950X turned out to be): of 47, 1 held, 22 held on the point estimate and 24 failed. Most speed predictions held (the fewest-admission set gains on Pf, and gains at least 0.03 over the greedy set at gpt-oss 11% on every machine; with two reads it is at most 0.01 behind the greedy set at 5 of 6 cells, and 0.047 behind on Pg at gpt-oss 25%); the other failures are the counters at gpt-oss 25% (copies 0.73 of greedy’s against at most 0.70, misses 5% more against at most 3%), the engine’s misses against the host’s replay, the two-read reads, and the interaction, which we predicted would shrink with the plan’s set and which grew (`prereg/minadm_outcome_104.md`).

Jobs 105–108: the time relation. Job 105 (`scripts/job105.py`; the plain form of Eq. (6) with G profiled on each host before its timed runs, the engine’s layer-ahead copy, and the fewest-admission set, on an EPYC 7402P, a Threadripper 9960X, Pf and O4): of 31, 7 held, 20 held on the point estimate and 4 failed. Every failure is the relation at gpt-oss 25%, which over-predicts the deployed cache by 6.9–9.3% (against at most 6%) on 3 of the four hosts (3.8% on Pf) and so takes the pooled median error (4.2%) past the 4% registered; at 11% it is within -1.5 to 4.7% . Every speed prediction held: the layer-ahead copy gains at least 3% on the host with a ratio above 0.8 and loses on both below 0.4, and the fewest-admission set beats the deployed cache on every host, beats the greedy set by at least 0.03 on the three with a ratio under 0.7, and copies at most 0.65 of the greedy set’s experts (`prereg/sumlaw_outcome_105.md`). Two clauses predicted quantities already measured, the profiled G (job 069c) and the plan’s copies, which are a property of the trace (job 104); they test that the engine reproduces them. Three more launched hosts produced no results: 105c never started its job, and the job’s bandwidth gate stopped 105d and 105g, whose cards were throttled, before any timed run. Job 106 (`scripts/job106.py`; Eq. (6) with its overlap term on both models at two budgets, the two online admission rules, and MIN’s sets launch by launch, on five hosts): of 181, 21 held, 112 held on the point estimate and 48 failed. 36 of the failures are on the two server machines new to the job, one of which computed wrong outputs and the other of which varied by more than the registered 2% between rounds (Appendix L); the three stable hosts are Pf, the Threadripper 9960X of job 105 and panel host Pe, all rented before. On the 3 stable hosts the relation held on every cell; the 11 failures there are the margin rule’s speed-up of at least 1.01 at gpt-oss on two hosts, the learned order’s speed-up of at least 1.00 on the fast-memory host, its host-time caps on the fastest-link host, and the relation’s prediction of both rules’ ratios on Pf at 25% (off by 0.030 and 0.045 against at most 0.03). The pooled clauses: the relation’s median error over all hosts failed (1), and the margin rule’s pooled median of at least 1.02 held only because of the two unstable machines (on the stable ones it is 1.01) (`prereg/onlineadmit_outcome_106.md`). Job 107 (`scripts/job107.py`; the same relation, the admission margin and MIN’s fewest-admission set loaded both

ways, on machines never rented before, with validity gates registered before launch): 7 machines rented, 3 stopped at the gate, 2 failed the round check, and the test had two valid machines of the three it required, which by its registered rule makes it inconclusive; its per-cell clauses failed on both. On them, of 46 clauses, 11 held, 22 held on the point estimate and 13 failed. The failures are the relation (7 of 8 cells beyond 6%, the pooled median error, and the overlap form against the plain one), the margin rule at one cell, the relation’s prediction of the margin rule’s speed at two cells on the server, and the count of valid machines. The profiled GPU compute, the margin rule’s reads and pooled median, both fewest-admission loads and the crossover between them held. Replacing machines that failed the round check, and the rules that chose the last two machines, were amendments committed before each machine started (`prereg/newhosts_outcome_107.md`). Job 108 (`scripts/job108.py`; the relation on machines never rented before, with the class drawn after job 107 gated before any download, gpt-oss only, and an 8% band): 6 machines rented, 2 stopped at the gate, 1 failed the round check, and three were valid. Of 21 clauses, 13 held on the point estimate, 6 failed and the rest could not be tested. The failures are the EPYC 7543’s two cells and its implied read rate at both budgets, and the two pooled clauses (2 of 6 cells within 6%, median error 6.7%); the profiled G of that host could not be tested (`prereg/smallhosts_outcome_108.md`).

The clause-by-clause tables (565 rows for jobs 073–098, 273 for job 099, 216 for job 100, 25 for job 101, 33 for job 102, 45 and 47 for jobs 103 and 104, 31 for job 105, 181 for job 106, 46 for job 107, 21 for job 108) are in the supplement (`paper/supplement.pdf`), generated by the same scripts.

D THE CALIBRATED TIME MODEL

Our cache’s one per-machine decision, how many of a layer’s misses to fetch rather than run on the CPU, comes from a calibrated model of time per token. With X_c the bytes the CPU reads per token and X_p the bytes the link reads,

$$T = G_{\text{fit}} + \max\left(\frac{X_c}{B_c}, \frac{X_p}{B_p}, \frac{X_c + X_p}{B_{cp}}\right), \quad (4)$$

with one constant G_{fit} per engine and model, fitted once on one host and frozen (it is neither the profiled G of Eq. (6) nor T_{GPU} of Eq. (2)). It is a calibrated model, not a law: G_{fit} is fitted, and it exceeds the whole all-in-VRAM token (5.17 against 3.92 ms) because it absorbs per-token host work and the CPU helpers’ shortfall against the probe (below). As a predictor of time it ties a naive one-path model with the same G_{fit} (time = G_{fit} + all host bytes over the best probed rate):

Held-out set	Eq. (4)		one path	
	median	90th pct.	median	90th pct.
blind test, 33 rows on 7 hosts	3.4%	12%	3.6%	31%
frozen, 30 rows on 4 later hosts	5.6%	16%	6.1%	11%

It over-predicts Qwen3’s speed on the later hosts by +5 to +21%, a budget-dependent bias whose G_{fit} was borrowed from llama.cpp. Its value is the per-machine fetch table it picks (Eq. (5) below), which gains 3.3–8.0% over a fixed table on host B and 9–34% on hosts with half the link (Appendix M).

The blind test in full. On every machine of the blind test the job wrote the model’s prediction from its own probe before any model ran (Table 9; all of its cache rows are gpt-oss-120b on RTX 5090 hosts). Over 29 configurations on 7 machines (33 measurements; one machine ran twice) the median error is 3.7% per configuration; without the 12-channel EPYC 9655 it is 3.3% over 24 configurations (90th percentile 7.5%). llama.cpp `--n-cpu-moe` is predicted within 4% on the Ryzen hosts, including two Qwen3 configurations on host B (+3.6%, +4.0%; for Qwen3, $G_{\text{fit}}=4.3$ ms, the mean residual of three llama.cpp runs on another host), and misses by +10–+119% on the hybrid-core and server hosts. With its constants frozen, the model scores the cache on the 4 RTX 5090 hosts rented later (30 measurements at the six cells of Table 4, not written before the runs) at a median 5.6%, 90th percentile 16%, worst 21% (Qwen3-30B-A3B 43.75% on oracle host O1). Refit under leave-one-host-out validation, the token-level max has a median error of 3.1% (90th percentile 9.4%), the additive form $G_{\text{fit}} + X_c/B_c + X_p/B_p$ 4.5% (11.3%) and a per-layer max summed over layers 3.2% (10.2%); the max beats the sum on the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, and on the EPYC 9655 rows, where both fail, the sum is closer on the two rows that copy behind a slow link, and elsewhere the two are within a few points (Appendix D). Each failure has a measured cause (Appendix D): on-demand copies behind a slow link share DRAM less evenly than the probe’s combined rate says; llama.cpp splits every matrix evenly across threads and runs at 46% of the model on a hybrid-core host; and on a 12-channel EPYC host memory stops being the limit and a per-layer

Table 9. The calibrated model predicted blind: each machine wrote its prediction from its own bandwidth probe, with constants frozen in the public repository, before any model run. Error of predicted over measured speed, gpt-oss-120b (cache rows) and llama.cpp --n-cpu-moe (last column, %).

Host CPU	Job	n	Median err %	Max err %	llama.cpp err %
Core Ultra 7 270K (8P+16E)	069c	5	1.5	11.9	+119
EPYC 7352 (Zen 2)	069c	5	3.2	11.7	+10
EPYC 9655 (12-ch. DDR5)	069c	5	18.7	31.9	+16
Ryzen 9 7900	072	5	4.9	8.2	+4
Ryzen 9 9950X3D (A)	073	8	1.6	4.5	—
Ryzen 9 7950X	074	1	4.0	4.0	—
Ryzen 9 9950X	076	4	2.5	5.0	—
Ryzen 9 9950X3D (B)	082	0	—	—	+4, +4

latency floor takes over. G_{fit} (5.17 ms refit) exceeds the whole all-in-VRAM token (3.92 ms) because it absorbs a shortfall the probe does not see: the helpers read at $1.09\text{--}1.31\times$ the probe’s time per expert, and a fitted rate factor is collinear with G_{fit} at this sample size.

What the model says, and the split it picks. Equation (4) is a calibrated model of execution that reads host memory in sequence with its GPU work, as our cache and llama.cpp do. For such an engine every mechanism acts only through the bytes read from host memory per token and the number of read paths in use: a second path adds bandwidth on every host of the blind test but the 12-channel EPYC (B_{cp} against $\max(B_c, B_p)$): 8–17% on its Ryzen hosts, 1% on the Core Ultra, 33% on the Zen 2 EPYC; the RTX 3090 host’s Core i9-11900KF, where the two paths together read less than the CPU alone, is a second exception), and that is the whole gain available to fetching or splitting misses between the CPU and PCIe. Prefetching never reduces the bytes; what it can buy is overlap of the reads with the step, which the model does not price: charging every read in sequence, it over-predicts the time per token of MIN prefetched by +1 to +38% (Section 5) and of a Belady prefetch by up to 53% (Appendix H). The model’s job in the system is the split. Its per-layer form,

$$T(n, f) = \max\left(\frac{fS}{B_p} + g, [n > f](L_c + \frac{(n-f)S}{B_c}), [0 < f < n] \frac{nS}{B_{cp}}\right), \quad (5)$$

with g the GPU’s expert work per layer ($48\mu\text{s}$ for Qwen3, $37\mu\text{s}$ for gpt-oss, from profiles) and $L_c=20\mu\text{s}$ the hand-off, picks for each count n of missed experts the number f to copy with the smallest T . On host B the model’s table beats the fixed table we used before at every budget of both models, by 3.3–8.0%; on hosts whose link we measured at 27–30 GB/s instead of 53 it fetches less and gains 21–34% on gpt-oss and 9–18% on Qwen3, the per-machine split mattering most where the link is slow (Appendix M). In the running example on host B the table is 0, 0, 1, 1, 2 ($f(0.4)$): one missed expert runs on the CPU, hidden behind the GPU’s other experts; from two misses the model copies one. The lead over FreeToken at that cell, $1.27\times$, is mostly the admission policy (77% of it by the LRU ablation of Appendix M), the rest the CPU path that FreeToken’s bandwidth model declines to use there.

Is the max the right form? In 19 of the 29 configurations nothing is copied, so they test one roof. We therefore refit the model and its alternatives on the same 33 measurements under leave-one-host-out validation (each of the 7 hosts predicted with constants fitted on the other six): the token-level max of Eq. (4) with G_{fit} refit has a median error of 3.1% (90th percentile 9.4%); the additive form $G_{\text{fit}} + X_c/B_c + X_p/B_p$, 4.5% (11.3%); a per-layer max summed over layers from the engine’s per-layer miss histograms, 3.2% (10.2%). The additive form ties in sample but misses the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, by -11%, while the max is within $\pm 0.2\%$

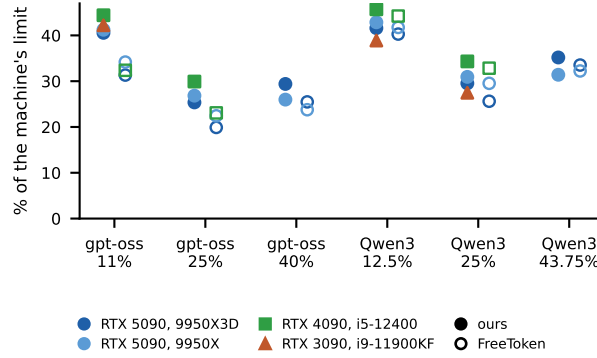


Figure 6. Ours (filled) and FreeToken (hollow) as a percentage of each machine’s own bound (exact optimum, the machine’s highest probed host rate, the card’s datasheet rate) at every cell of Table 10: two RTX 5090 hosts, an RTX 4090 and an RTX 3090 (FreeToken’s 3090 cells, its carried-over backend running below llama.cpp, are omitted). The limit ranges from 41 to 519 tok/s across these cells; ours stays at 25–46% of it.

there; on the slow-link copy rows the additive form is the closer of the two. The per-layer form neither beats the max nor explains its failures: the Jensen gap between a sum of per-layer maxima and the maximum of sums is a fifth of the Core Ultra’s residual, none of the EPYC 7352’s, and has the wrong sign on the EPYC 9655 and on the prefetch rows. We keep Eq. (4), with its evidence stated: the max beats the sum on the three fast-link copy rows and on the EPYC 9655 rows, where both fail, the sum is closer on the two slow-link copy rows, and elsewhere the two are within a few points, so the form is settled only where both paths carry most of the bytes. FreeToken’s own runs are not predicted, because its counters are not exposed; the model covers our engine and llama.cpp.

Where it fails, and why. Each failure now has a measured cause. (i) Copying misses on demand (FETCH) at 25% is over-predicted by +11.7% on the Core Ultra 7 and +11.4% on the EPYC 7352 (links of 36 and 26 GB/s); +8.8% and +8.5% with G_{fit} refit. The engine’s per-layer timers show why: a CPU miss served beside a copy takes 20–62% longer than one served alone, so the two paths share DRAM less evenly than the probe’s combined rate says; a variant of the model that uses the probe’s per-path rates under contention and a per-layer hand-off constant predicts both rows within +3.1% and +4.1%, at a worse median (4.2%). (ii) llama.cpp on a Core Ultra 7 with 8 performance and 16 efficiency cores runs at 46% of the model’s prediction at its default thread count, because it splits every matrix evenly across its threads; our helpers claim work dynamically and stay on the model on the same machine. We did not test llama.cpp restricted to the performance cores. (iii) On a 12-channel EPYC 9655 (559 GB/s) host memory stops being the limit (+16 to +29% with G_{fit} refit under leave-one-host-out validation; Table 9 gives the blind errors): the per-layer timers show a floor of 55–70 μs per layer with CPU misses that no desktop host has (their intercepts are at or below zero), the latency of the hand-off at a bandwidth where a 13 MB expert takes 24 μs . (iv) Prefetch configurations are under-predicted by -8.7 to -3.4%: their copies overlap attention, which the model charges in full.

E THE GRID: OTHER CARDS, AND THE SCOPE OF THE GAIN

Table 10 lists every cell of Table 4’s protocol on the four machines we measured, each against its own limit, and Fig. 6 plots the fractions. The protocol was repeated on an RTX 4090 next to a Core i5-12400 (the host reads 37 GB/s, the link 25) and an RTX 3090 next to a Core i9-11900KF (45 GB/s; CPU and link together read *less* than the CPU alone, so the model’s table copies nothing), at the cells that fit 24 GB. On the 4090 ours runs $1.03\text{--}1.37\times$ FreeToken and $1.8\text{--}2.6\times$ llama.cpp at half host B’s absolute speeds, the host being the reason; on the 3090 it runs $1.6\text{--}2.2\times$ llama.cpp (FreeToken’s carried-over backend ran below llama.cpp there and is not counted). The cards’ own bandwidth never enters: every one of the seven cells on the 24 GB cards is host-bound at the limit, and ours stands at 30–46% and 27–42% of it. Mixtral-8x7B (8 experts per layer, top-2; 26 GB of experts on the 3090) marks the scope: with 2 of 8 experts resident the cache hits 31% of reads where whole-layer pinning hits 25%, so there is little locality to earn, and ours ties llama.cpp (1.015 [1.008, 1.024]); on the 128-expert models the same cache hits 53–79% at 11–25% resident. Across eight hosts the gpt-oss lead over FreeToken grows with the host’s CPU-to-link ratio, from $1.09\text{--}1.22\times$ where the CPU reads $1.05\text{--}1.14\times$ its link’s rate to $1.51\text{--}1.58\times$ behind a link of half the bandwidth, in the direction the model’s split implies (the model does not predict FreeToken’s runs),

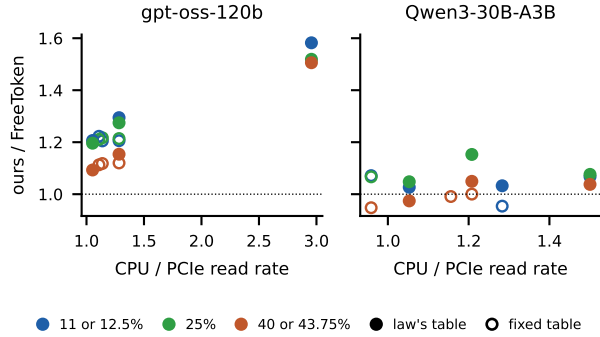


Figure 7. Ours over FreeToken’s better backend (carried over from host B on host S) in every same-machine comparison of the current engine on the two models of Table 4 (eight hosts), against the machine’s CPU-to-PCIe read ratio (FreeToken’s own probe). Filled: FETCH table from the model on that machine; hollow: the fixed table. The rightmost gpt-oss points are a Core i9-14900K with FreeToken’s executor on its performance cores; the rightmost Qwen3 points are a Ryzen 9 7900.

and the Qwen3 lead stays at $0.95\text{--}1.15\times$ (Fig. 7, Appendix M).

The 24 GB cards run the cells that fit (the VRAM arithmetic, written on the machine from the GGUF header, skipped Qwen3 at 43.75%: 28.0 GB of slots, dense weights and KV against a 23 GB limit). The predictions for jobs 091 and 092 and their outcomes are in `prereg/grid_outcome_091.md` and `092.md`: on the 4090 our lead over FreeToken at gpt-oss 11% (37%) exceeded the predicted 10–35% band, ours ran $1.78\times$ llama.cpp at Qwen3 12.5% against a predicted 1.8, the model’s tables copied one more expert than host B’s at one entry each (the prediction reasoned from the link alone; the CPU is slower in the same proportion), and ours’ fraction of the limit at Qwen3 12.5% (45.7%) exceeded the predicted 25–45%; on the 3090 FreeToken’s carried-over backend ran below llama.cpp, ours ran $1.56\text{--}1.64\times$ llama.cpp at two cells against a predicted 1.8, Mixtral tied at $C=2$ against a predicted $1.3\times$ and its $C=4$ cell was lost to the job’s deadline, and one launch-order check exceeded 3% (7%).

F THE ORDER-FREE ACCOUNTING OF THE GAP

Two choices separate the deployed cache from MIN with one read: *what to cache* (the deployed policy’s admissions or MIN’s) and *how to load* (by the CPU, two reads, or in the step, one). The engine ran all four combinations on O4, O5, the panel and the three new machines of jobs 100–101. In the running example, changing only what to cache closes 4.5% of the 10.6 ms gap, changing only how to load 3.9%, and both 51.5% (rounded): the choices interact, because MIN caches 4.6 times as many experts per token as the deployed policy and, loaded by the CPU, each is read twice. We credit each choice with its effect averaged over the two orders, its Shapley value (Fawcett & Hoos, 2016; Jain, 1991) (26.1% and 25.5% here); Table 11 also lists each alone and the interaction. Copying ahead is nested under MIN’s set (12.7%); what no oracle touches is the rest (35.7%). On the 13 hosts whose link reads at least half as fast as the CPU, either change alone closes at most 26% of the gap and both together 14–60%; on the 2 below that line, loading in the step costs time at 3 of 4 host-budgets, and at 3 of 4 on 2 more (job 103). These use the greedy set. With the fewest-admission set (job 104), both together close 6–49% at 11% on its three machines, against $-26\text{--}39\%$ with the greedy set, and the interaction remains (1–33%).

G THE ACCOUNTING AS A MODEL

Between the measured time of a cell and its limit lie five modelled fixes: overlap of host reads with GPU work (in Eq. (1)’s form, which also rebalances experts onto the CPU), the optimum’s reads instead of the online policy’s (*foresight*), the best online policy’s reads at the probe’s highest rate instead of the engine’s through Eq. (4) (policy and read paths), the engine’s per-token host work, and GPU kernels at the datasheet rate. Adding them in a fixed order makes the first one added look largest, so we report each fix’s Shapley value, its average marginal cost over every order (Fig. 8), a method with a precedent in performance debugging of microservices (Li et al., 2024). Two of the five terms are definitions rather than measurements: the GPU term is the residual, its factor calibrated per cell so that the model with no fix applied reproduces the measured time, and “overlap” is Eq. (1)’s form. On host B the model has two regimes. In the four host-bound cells missing foresight is the largest cost, 37–53% of the gap (26–31% of the token; in the running example 37%, where it edges overlap by 0.14 ms

Table 10. The grid: Table 4’s protocol on every card and host we measured, each cell against its own machine’s limit (the exact optimum’s reads on the same trace, the machine’s highest probed host rate, the card’s datasheet rate). *Bound*: what binds the limit; *host* when the optimum’s reads alone take longer than the GPU’s whole read, so the GPU has slack, *both* when the limit runs experts on the CPU until the two paths take equally long, which is where the all-in-VRAM speed can be exceeded. Ours and FreeToken are launch 2 (FreeToken’s backend carried over from host B’s selection); the 40 and 43.75% budgets do not fit a 24 GB card. [†]On the RTX 3090 FreeToken’s carried-over hybrid backend ran below llama.cpp; its own calibration recommends its offload backend there, which we did not run, so its speed is reported and no ratio against it is printed. Mixtral-8x7B (26 GB of experts, top-2 of 8) is ours against llama.cpp only, on the RTX 3090, launch 1.

Model	Experts on GPU	llama.cpp (tok/s)	FreeToken (tok/s)	Ours (tok/s)	Ours ÷ FreeToken (95% CI)	Ours ÷ llama.cpp	Limit (tok/s)	Bound	Ours, % of limit
RTX 5090 , 9950X3D (host B): B_{host} 88 GB/s, B_{gpu} 1792 GB/s; tables 0, 0, 1, 1, 2 / 0, 0, 1, 1, 2, 3, 3, 4, 5									
gpt-oss-120b	11%	34.9	54.0	69.9	1.29 [1.28, 1.31]	2.00	172	host	41
gpt-oss-120b	25%	40.0	85.6	109.2	1.27 [1.25, 1.29]	2.73	430	host	25
gpt-oss-120b	40%	47.7	132.2	152.5	1.15 [1.13, 1.17]	3.20	519	both	29
Qwen3-30B-A3B	12.5%	19.5	38.7	40.0	1.03 [1.02, 1.04]	2.05	96	host	42
Qwen3-30B-A3B	25%	22.1	54.7	63.1	1.15 [1.13, 1.17]	2.86	214	host	30
Qwen3-30B-A3B	43.75%	28.4	103.1	108.2	1.05 [1.04, 1.06]	3.81	308	both	35
RTX 5090 , 9950X (host S): B_{host} 71 GB/s, B_{gpu} 1792 GB/s; tables 0, 0, 1, 2, 3 / 0, 0, 1, 2, 2, 3, 4, 5, 6									
gpt-oss-120b	11%	28.9	48.0	57.9	1.21 [1.19, 1.22]	2.00	140	host	41
gpt-oss-120b	25%	33.6	78.7	94.1	1.20 [1.16, 1.22]	2.80	351	host	27
gpt-oss-120b	40%	40.1	122.2	133.6	1.09 [1.08, 1.11]	3.34	514	both	26
Qwen3-30B-A3B	12.5%	16.1	32.7	33.5	1.03 [1.02, 1.04]	2.08	78	host	43
Qwen3-30B-A3B	25%	18.5	51.4	53.9	1.05 [1.04, 1.06]	2.91	174	host	31
Qwen3-30B-A3B	43.75%	23.8	98.3	95.8	0.97 [0.96, 0.99]	4.02	305	both	31
RTX 4090 , i5-12400 (job 091): B_{host} 37 GB/s, B_{gpu} 1008 GB/s; tables 0, 0, 1, 2, 2 / 0, 0, 1, 1, 2, 3, 4, 4, 5									
gpt-oss-120b	11%	18.1	23.8	32.7	1.37 [1.35, 1.39]	1.81	74	host	44
gpt-oss-120b	25%	21.0	42.5	55.0	1.30 [1.26, 1.33]	2.62	184	host	30
Qwen3-30B-A3B	12.5%	10.5	18.2	18.8	1.03 [1.02, 1.05]	1.78	41	host	46
Qwen3-30B-A3B	25%	12.3	30.0	31.3	1.05 [1.03, 1.06]	2.54	91	host	34
RTX 3090 , i9-11900KF (job 092): B_{host} 45 GB/s, B_{gpu} 936 GB/s; tables 0, 0, 0, 0, 0 / 0, 0, 0, 0, 0, 0, 0, 0, 0									
gpt-oss-120b	11%	22.8	8.4 [†]	37.3	–	1.64	88	host	42
Qwen3-30B-A3B	12.5%	12.3	9.0 [†]	19.2	–	1.56	49	host	39
Qwen3-30B-A3B	25%	13.8	11.2 [†]	30.1	–	2.18	110	host	27
Mixtral-8x7B Q4_K_M	$C=2$ (25%)	8.22	–	8.34	–	1.015 [1.008, 1.024]	–	–	–

and is the largest marginal cost in 70 of the 120 orders); in the two GPU-bound cells host reads serialised with GPU work is largest, 39–46% of the gap, and foresight second at 23%. Policy and read paths cost 12–16%, per-token host work 3–10%, and GPU kernels below datasheet, net of the overlap the engine achieves, 4–16%. The regimes are a property of the host: recomputed for the three oracle hosts, whose host memory is weaker, foresight edges overlap at three of their six GPU-bound cells. The model prices foresight and overlap as nearly independent players, and the interior states it values are not realised by any online policy. The rest of this section measures them.

H FORESIGHT IN THE ENGINE

Four jobs ran oracle policies in the engine, each on a rented RTX 5090 host with weaker host memory than host B’s. Jobs 093–097 report ratios of mean speeds rather than of mean times per token; recomputed either way, no ratio moves by more than 0.006. The fourth (096) is the confirmatory factorial of Fig. 3: ten configurations per cell, each in a fresh context, in an order shuffled per cell with a recorded seed, on hosts O4 (Ryzen 9 9950X: 43, 45 and 51 GB/s by the CPU, over the link and at best) and O5 (Ryzen 9 9950X3D: 47, 27 and 51 GB/s), with ten predictions per host committed before either started (prereg/foresight_outcome_096.md). They held at most cells. The failures: fetch’s gain above the predicted band at Qwen3 25% on O4 and different between the hosts by up to 0.26 against the 0.15 predicted; Belady’s single-read prefetch faster than predicted at Qwen3 12.5% on O4, with reads below the predicted 1.25–1.9× the MIN prefetch’s at Qwen3 12.5% on both hosts and at gpt-oss 11% on O4 (down to 1.09×); the bytes share at Qwen3 12.5% on O4 (60% of the gap against a predicted 25–55%); and the reads fetch-on-admit saves at gpt-oss 40% on O5, just above the predicted 22%. Reading the online policy’s admissions once came to 0.97–1.03× at the host-bound cells, at the bottom of the predicted 0.97–1.12× although its reads fell as predicted, and to 0.95–1.02× at the GPU-bound cells. Table 12 lists every cell of the single-read

Table 11. An order-free accounting of the gap between the deployed cache and the bound at the host-bound budgets, as percentages of the gap (ms per token). The engine ran both choices of what to cache (the deployed policy’s admissions or MIN’s) with both ways to load a cached expert (two reads: the CPU serves it, then a copy; one read: a copy in the step). *Load alone* and *cache alone* are each choice’s effect with the other at the deployed level; the *interaction* is how much more MIN’s set is worth loaded once than loaded twice. The Shapley values average each choice’s effect over the other’s two levels and add up, with the prefetch (nested under MIN’s set, since it needs foresight) and the rest, to 100%. Panel: mean over its hosts (number in parentheses) and, in small type, the range over its hosts (not an interval); a fast link reads at least half as fast as the host’s CPU, a slow one less. Last rows: the new machines of the follow-up launches, with their link-to-CPU ratio.

Host	Budget	Gap (ms)	Load alone	Cache alone	Interaction	Shapley value cache	load	Prefetch	Rest
O4	gpt-oss 11%	10.6	4	4	43	26	25	13	36
O4	gpt-oss 25%	8.5	4	22	17	31	13	22	34
O4	Qwen3 12.5%	18.4	3	−1	58	28	32	4	36
O4	Qwen3 25%	14.1	2	14	37	33	21	17	30
O5	gpt-oss 11%	11.4	0	−2	28	12	14	17	57
O5	gpt-oss 25%	9.1	−4	20	10	25	1	21	52
O5	Qwen3 12.5%	19.8	1	−13	43	9	23	7	62
O5	Qwen3 25%	15.3	0	14	17	22	9	16	53
Panel, fast link (8)	gpt-oss 11%	11.6	1 (−2 to 4)	3 (−2 to 6)	38 (30 to 47)	22 (14 to 26)	20 (14 to 27)	13 (−4 to 20)	45 (38 to 55)
Panel, fast link (8)	gpt-oss 25%	9.0	0 (−4 to 3)	22 (18 to 26)	15 (10 to 19)	29 (25 to 35)	8 (1 to 12)	20 (9 to 23)	43 (38 to 49)
Panel, slow link (2)	gpt-oss 11%	9.7	−6 (−9 to −3)	−4 (−9 to 0)	3 (−6 to 11)	−3 (−11 to 6)	−5 (−12 to 2)	17 (9 to 24)	91 (83 to 99)
Panel, slow link (2)	gpt-oss 25%	7.8	−17 (−22 to −11)	19 (16 to 23)	3 (3 to 3)	21 (17 to 25)	−15 (−21 to −10)	22 (17 to 26)	73 (68 to 78)
5700X3D (0.92)	gpt-oss 11%	14.9	4	−2	55	25	31	8	36
5700X3D (0.92)	gpt-oss 25%	11.5	4	22	24	34	17	18	31
9950X, x8 (0.54)	gpt-oss 11%	8.3	0	−3	17	5	8	28	59
9950X, x8 (0.54)	gpt-oss 25%	7.4	−15	13	17	22	−6	26	59
9800X3D, slower link (0.61)	gpt-oss 11%	10.8	−2	−4	29	11	12	20	57
9800X3D, slower link (0.61)	gpt-oss 25%	8.9	−5	20	12	26	1	20	54

factorial of Section 5 on host O3 (a Ryzen 9 9950X whose probe reads 53 GB/s by the CPU at the helper count, 43 over the link, 60 with both and 61.5 at best; the model’s FETCH tables 0, 0, 1, 2, 2 for gpt-oss and 0, 0, 1, 1, 2, 3, 4, 4, 5 for Qwen3), and Table 13 the milliseconds per token of the engine states the three jobs realised beside each host’s limit and the model’s values. Table 14 lists the first job (093), the hit-optimal oracle with a window sweep on host O1 (a Ryzen 9 9950X: 44, 46, 49 and 50.5 GB/s; tables 0, 0, 1, 2, 3 and 0, 0, 1, 2, 3, 4, 5, 6, 7), and Table 15 the second (094), MIN with bypass with background admissions beside the hit-optimal oracle on host O2 (a Ryzen 9 7950X: 44, 46, 50 and 50.6 GB/s; the same tables). On O1 and O2 every cell but gpt-oss 40% and Qwen3 43.75% is host-bound at the limit; on O3, whose memory is stronger, the same four.

The single-read oracles. In the fetch oracle the step’s plan is computed before the layer runs from the trace’s record of the current step: the misses MIN with bypass would admit are fetched by the engine’s FETCH operation into their victims’ slots and run from there, and every other miss runs on the CPU; the oracle refuses an admission when no resident’s next use is further than the candidate’s. The single-read prefetch admits, each step, the non-resident experts whose first use is at least d steps away ($d=2$ unpaced, 3 paced), provided the victim’s next use is further than the candidate’s use after that first one; an expert used sooner than d steps away misses on the CPU, or is fetched when the two are combined. The same policies simulated on the trace with the engine’s semantics (the lookahead within the sequence, residents carried over with their next use recomputed at each sequence start) predict the engine’s reads and hit rates: the fetch oracle’s hit rate within 0.4 points and its reads within 0 to 4%, the combined oracle’s hits within two points (Table 12). The simulator the job’s predictions were registered against gave a resident carried over from the previous sequence no next use, so it was evicted first; that over-counted the fetch oracle’s reads by 17% at the two GPU-bound cells, which is why two of its reads clauses failed (Appendix C). Each configuration of llama-ec-bench starts with a cold cache in a fresh context. The paced copy path copies one 16 MB piece at a time on a single copier thread and publishes a landed copy at the start of the next step; the unpaced path copies each admission as one transfer. The hit-optimal oracle’s background admissions drive host memory at 71–82% of the probe’s best rate at the host-bound cells, the prefetching single-read oracles at 64–85% (the paced one 71–82%), the online policy itself at 66–82%, the fetch oracle at 42–67%. The predictions and their outcomes are in prereg/foresight_outcome.095.md: of 56 clauses, 44 held and 12 failed, all of size (Appendix C).

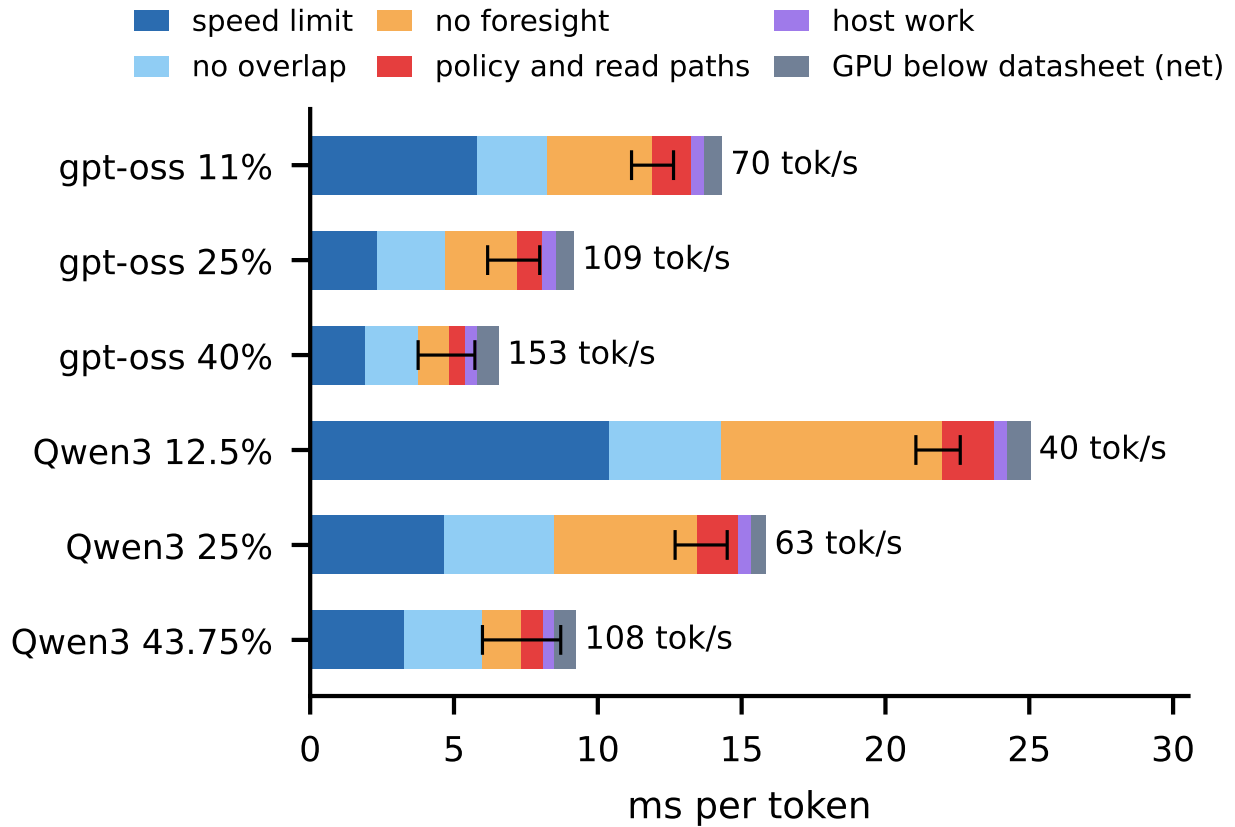


Figure 8. The accounting as a model: Shapley attribution of the gap between our cache’s measured time per token and the bound, for every cell of Table 4 on host B, averaged over all 120 orders in which the five fixes can be applied. Fixes: overlapping host reads with GPU work, which in Eq. (1)’s form also moves experts onto the CPU until the two paths balance; the optimum’s reads instead of the best online policy’s (foresight); the best online policy’s reads at the probe’s highest rate instead of the engine’s reads through Eq. (4) (policy and read paths); the engine’s per-token host work; and GPU kernels at the datasheet rate, the term that closes the model on the measured time. Whiskers: the range of foresight’s marginal cost over the orders.

The double-reading oracles. Both read the routing of the next W decode steps; their copies ride the paced admission path, and an expert still in flight at its step runs on the CPU like any other miss. The *hit-optimal* oracle admits the experts those steps will route to, soonest first, evicting the resident whose next use is furthest (Belady within the window), for one future use as readily as for ten. The *bytes-optimal* oracle is MIN with bypass with its admissions made through the background path: a miss is served where it is, by the CPU or a demand copy, and admitted only if its next use comes before the furthest next use among the residents. With the whole sequence in view the hit-optimal oracle runs $1.40\times [1.38, 1.41]$ the online policy at gpt-oss 25% and $1.25\times [1.24, 1.26]$ at Qwen3 25% on host O1, $1.63\times$ and $1.41\times$ on O2, and $1.48\text{--}1.62\times$ at the two GPU-bound cells; half of its gain at gpt-oss 25% arrives between $W=4$ and 16 (9.2 tokens by log interpolation; Qwen3 25%: 6.1 against the trace’s 3.9). Where the host binds hardest it loses, 3–8% at gpt-oss 11% and Qwen3 12.5% on both hosts, and at every window on O1 ($0.78\text{--}0.92\times$): it reads more host bytes than the online policy (82 and 195 experts per token against 63 and 160), $2.02\text{--}2.18\times$ the optimum’s R^* , because Belady admits an expert used once where the optimum bypasses it and re-copies what it evicts (its admissions, each one read, are most of its reads; a miss served on the CPU before its copy lands is the second read of that expert). An unpaced copy path, run at the four host-bound cells of O1, reaches 98–99% hits at three of them and still loses at gpt-oss 11% and Qwen3 12.5% (where it hits 75%), so copies that arrive in time do not remove the loss there: the extra bytes cause it. The bytes-optimal oracle gains 13–23% at the four cells

Table 12. The single-read oracles (job 095; RTX 5090, Ryzen 9 9950X, host memory 61.5 GB/s at best) beside the hit-optimal one. Per cell: the online policy’s speed and its share of this host’s limit; for each oracle with the whole sequence in view, its ratio to the online policy (paired, 95% interval), hit rate, host reads per token (CPU misses + fetches + admissions) as a multiple of the optimum’s (each read counted once), share of the limit and the calibrated model on its own counters against its measured time. *fetch*: MIN with bypass, admitted misses fetched into their slot this step; *lead*: the scheduled single-read prefetch (first use at least two steps away); *both*: fetch and lead; *paced*: both on the paced copy path with a three-step lead; *hit-opt.*: the hit-optimal oracle (Belady prefetch without bypass, unpaced). The simulation columns give the trace’s prediction of reads and hits for fetch and both.

Cell	online			fetch				both				hit-opt.				lead		paced		simulated			
	tok/s	% lim.	ratio	hits	reads/R*	% lim.	model	ratio	hits	reads/R*	% lim.	ratio	hits	reads/R*	% lim.	ratio	% lim.	ratio	% lim.	fetch reads	hits	both reads	hits
gpt-oss 11%	55.5	46	1.29 [1.28, 1.29]	72	1.04	59	-8% 1.39 [1.39, 1.40]	86	1.23	64	0.91 [0.90, 0.92]	98	1.77	42	1.26	58	1.45	67	39	73	46	87	
gpt-oss 25%	90.3	30	1.32 [1.31, 1.33]	89	1.08	39	-13% 1.49 [1.48, 1.50]	97	1.45	44	1.14 [1.12, 1.15]	99	2.11	34	1.45	43	1.66	50	16	89	22	99	
gpt-oss 40%	128.8	25	1.25 [1.24, 1.27]	95	1.14	32	-14% 1.39 [1.37, 1.41]	99	1.60	35	1.27 [1.25, 1.28]	99	2.20	32	1.38	35	1.49	38	7	95	12	100	
Qwen3 12.5%	32.3	48	1.34 [1.34, 1.35]	74	1.04	64	-9% 1.42 [1.41, 1.43]	84	1.20	68	0.69 [0.68, 0.71]	77	2.47	33	1.22	58	1.41	68	100	74	115	84	
Qwen3 25%	51.8	34	1.38 [1.37, 1.39]	88	1.04	48	-15% 1.48 [1.46, 1.49]	96	1.42	51	1.06 [1.05, 1.08]	99	2.00	37	1.41	48	1.67	57	45	88	61	97	
Qwen3 43.75%	91.4	30	1.26 [1.25, 1.28]	96	1.08	38	-17% 1.41 [1.39, 1.42]	99	1.62	42	1.27 [1.26, 1.29]	100	2.29	38	1.40	42	1.50	45	15	96	23	100	

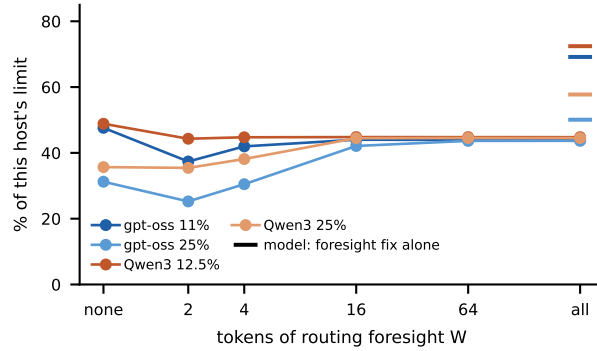


Figure 9. The first oracle job (093, host O1): speed as a percentage of this host’s limit against the window W of future routing the hit-optimal oracle sees, at the four host-bound cells. “none” is the online policy; “all” the whole sequence; points are connected for the trend only. The tick at the right is the model’s foresight-only term for the cell, the speed it predicts with that one fix; the oracle’s host traffic sits at 89–94% of the probe’s best rate at every window, so its speed is R^* over its reads times that utilisation, 44–45% of the limit with the whole sequence in view.

of 25% and above and 0–3% at the two below (6% with a 16-token window); its misses fall (30 to 20 and 85 to 64 per token at 25%) but its total reads, misses plus the admissions that read each admitted expert again, are $1.73\text{--}2.04 \times R^*$ and exceed the online policy’s by 4–5% at the two lowest budgets; the model on its own counters predicts its time within 9% at every cell, and the bytes it saves account for 4–21 points of its 13–23% gain at the four cells (most of it at gpt-oss 40% and Qwen3 43.75%, a third at Qwen3 25%). The hit-optimal oracle reads $1.01\text{--}1.13 \times$ the bytes-optimal one’s host bytes on O2 and gains 41–63% at the same four cells, and the model, with its GPU term calibrated on the online run, over-predicts those runs’ time per token by 37–53% (on O1, 24–39%); its gain is the overlap of copies issued ahead of their use with the step, which the model charges in sequence. Pacing the online policy’s own copies changes it by under 1% ($0.997\text{--}1.005 \times$). With no FETCH table (every miss on the CPU) the online policy runs 8–15% slower at the four host-bound cells; with the engine’s within-layer overlap of a layer’s CPU misses with its GPU hits turned off it changes by under 1% at every cell (0.5% faster to 0.6% slower), so the overlap the accounting charges is across the step and in rebalancing, not within a layer. The bytes-optimal oracle with a 16-token window keeps 34–176% of the whole-sequence gain at the five cells where that gain’s interval excludes zero (more than all of it at gpt-oss 11%, where the shorter window admits less).

Why the bytes-optimal oracle’s misses stay above the optimum’s reads. Besides the second read, its misses alone are $1.31\text{--}1.47 \times$ the optimum’s reads at the four host-bound cells (at the two GPU-bound ones the ideal simulation below is itself 6–10% above the optimum, which sees across sequences). The engine’s paced copy path moves one 16 MB piece at a time from pageable host memory on a single copier thread, behind the demand copies, publishes a landed copy at the start of the next step, and admits nothing while 64 copies are queued; at the 16–30 admissions per token (210–285 MB) of the two lowest budgets it lags the admissions by steps, and an expert still in flight is a miss the CPU reads again. The last four columns of Table 15 simulate MIN with bypass on the measured routing trace with the engine’s own semantics (the lookahead within

Table 13. The accounting, measured: milliseconds per token of the engine’s states that the oracle jobs realised, on each oracle host, beside that host’s limit and the model’s values for the states it prices; O1–O3 ran jobs 093–095 in a fixed order, O4 and O5 job 096 in a shuffled order. *online*: the deployed policy; *foa*: the same with each admission fetched into its slot in the step that misses it (one read, no foresight); *no ovl.*: its within-layer overlap off; *all CPU*: no FETCH table; *bypass*: MIN with bypass with background admissions (two reads per admitted expert); *hit-opt.*: Belady prefetch without bypass, which admits what the optimum bypasses, unpaced, and the same on the paced copy path (the first *paced* column; O1 and O2 ran only that form); *nb2*: Belady within a two-step lead with each admission read once; *fetch*: MIN with bypass, admitted misses fetched into their slot (one read, serialised: the model’s foresight-only state $v(F)$); *both*: fetch and the scheduled single-read prefetch (one read, overlapped: the model’s $v(O, F)$); the second *paced*: both on the paced copy path with a three-step lead; *limit*: the limit on that host; $v(F)$, $v(O, F)$, $v(\text{all})$: the model.

Host	Cell	online	foa	no ovl.	all CPU	bypass	hit-opt.	paced	nb2	fetch	both	paced	limit	$v(F)$	$v(O, F)$	$v(\text{all})$
O1	gpt-oss 11%	21.12	–	21.01	22.91	–	–	22.88	–	–	–	–	10.06	14.54	11.27	10.06
O1	gpt-oss 25%	12.89	–	12.91	14.21	–	–	9.22	–	–	–	–	4.03	8.04	5.10	4.03
O1	gpt-oss 40%	8.90	–	8.95	–	–	–	5.99	–	–	–	–	1.97	5.75	3.84	1.97
O1	Qwen3 12.5%	36.92	–	36.91	43.29	–	–	40.29	–	–	–	–	18.04	24.90	19.29	18.04
O1	Qwen3 25%	22.73	–	22.76	26.60	–	–	18.22	–	–	–	–	8.11	14.04	9.22	8.11
O1	Qwen3 43.75%	12.71	–	12.67	–	–	–	8.12	–	–	–	–	3.32	8.25	5.51	3.32
O2	gpt-oss 11%	20.85	–	–	–	20.17	–	21.57	–	–	–	–	10.04	14.27	11.96	10.04
O2	gpt-oss 25%	12.72	–	–	–	10.77	–	7.80	–	–	–	–	4.02	7.86	5.86	4.02
O2	gpt-oss 40%	8.79	–	–	–	7.13	–	5.62	–	–	–	–	1.97	5.63	3.78	1.97
O2	Qwen3 12.5%	36.64	–	–	–	36.58	–	38.74	–	–	–	–	18.00	24.70	20.06	18.00
O2	Qwen3 25%	22.24	–	–	–	19.67	–	15.80	–	–	–	–	8.09	13.58	9.90	8.09
O2	Qwen3 43.75%	12.02	–	–	–	10.11	–	7.42	–	–	–	–	3.32	7.56	4.90	3.32
O3	gpt-oss 11%	18.01	–	–	–	–	19.79	–	–	14.00	12.91	12.39	8.26	12.63	9.49	8.26
O3	gpt-oss 25%	11.08	–	–	–	–	9.74	–	–	8.42	7.44	6.66	3.31	7.05	4.28	3.31
O3	gpt-oss 40%	7.77	–	–	–	–	6.14	–	–	6.20	5.58	5.20	1.95	5.24	3.56	1.95
O3	Qwen3 12.5%	30.95	–	–	–	–	44.54	–	–	23.02	21.78	21.89	14.81	20.35	16.06	14.81
O3	Qwen3 25%	19.32	–	–	–	–	18.17	–	–	14.01	13.09	11.58	6.66	11.68	7.76	6.66
O3	Qwen3 43.75%	10.94	–	–	–	–	8.59	–	–	8.66	7.78	7.28	3.30	7.24	4.76	3.30
O4	gpt-oss 11%	20.59	20.18	–	–	20.11	19.78	22.03	19.64	15.13	14.06	13.78	10.02	13.77	11.63	10.02
O4	gpt-oss 25%	12.43	12.07	–	–	10.55	9.76	8.13	9.61	8.72	7.65	6.87	4.01	7.40	5.34	4.01
O4	gpt-oss 40%	8.39	8.19	–	–	6.91	6.15	5.38	6.03	6.28	5.59	5.21	1.97	5.12	3.20	1.97
O4	Qwen3 12.5%	36.36	35.83	–	–	36.54	48.80	39.66	31.64	25.29	24.01	24.58	17.97	23.98	19.88	17.97
O4	Qwen3 25%	22.09	21.75	–	–	20.06	17.74	16.93	17.57	14.60	13.50	12.24	8.08	13.12	9.54	8.08
O4	Qwen3 43.75%	11.94	11.74	–	–	10.09	8.44	7.17	8.33	8.69	7.72	7.29	3.32	7.32	4.56	3.32
O5	gpt-oss 11%	21.26	21.31	–	–	21.47	33.34	25.57	33.03	18.34	17.97	16.39	9.94	15.22	11.20	9.94
O5	gpt-oss 25%	13.01	13.35	–	–	11.18	15.87	11.17	15.60	10.60	10.54	8.68	3.98	9.13	5.06	3.98
O5	gpt-oss 40%	8.73	9.22	–	–	7.16	8.34	6.25	8.13	7.27	6.87	5.91	1.97	6.26	3.62	1.97
O5	Qwen3 12.5%	37.63	37.42	–	–	40.11	69.42	44.89	49.11	31.37	31.27	30.01	17.82	27.95	19.27	17.82
O5	Qwen3 25%	23.24	23.27	–	–	21.12	30.78	21.76	30.47	18.51	19.64	16.03	8.01	16.77	9.15	8.01
O5	Qwen3 43.75%	12.49	12.92	–	–	10.32	12.08	9.00	11.86	10.21	9.61	8.23	3.31	8.94	5.02	3.31

the sequence, residents carried over, no queue cap) and a copy that lands d steps after it is issued, during which the expert misses and is not re-admitted: at $d=1$, the ideal, the misses come within 0–4% of the limit’s optimum at the host-bound cells; the engine’s misses meet the simulation at $d=2.8$ – 4.7 there and at 3.0–3.5 at the GPU-bound cells, three to five steps. At the fitted latency the simulation admits 1.1 – $1.9\times$ what the engine admitted, so the queue cap, not only the latency, keeps the engine’s misses above the ideal. The predictions and their outcomes are in `prereg/foresight_outcome_093.md` and `094.md`: of 36 clauses for the first job, 9 held, 2 held on the point estimate, 21 failed and 4 were untested; of 38 for the second, 12 held, 1 on the point estimate, 24 failed and 1 was untested (Appendix C). The second job’s failures are of size, not direction: the bypass oracle’s misses were predicted within 15% of the optimum’s reads and came 29–47% above (the copy latency and the queue cap); its hit rate was predicted within 4 points of the optimum’s and fell 5–11 points short at three cells; its gain was predicted at 10–45% at the host-bound cells and came below that at the two lowest budgets; the model was predicted to over-predict it by 10–40% at the host-bound cells and over-predicted by 5–9%; and the two oracles were predicted within 15% of each other at 25% and above, where the hit-optimal one was 1.24 – $1.38\times$ faster.

Where a token’s time goes at a GPU-bound budget. On Qwen3 at 43.75% Nsight Systems splits a token of our cache (FreeToken in parentheses) into 4.93 (5.20) ms of expert copies over PCIe, 2.29 (2.21) of resident-expert products, 1.65 (1.34) of attention projections, 0.49 (0.37) of attention, 0.38 of output head and 0.97 (0.40) of norms, small kernels and cache control, with neither system overlapping its copies with GPU compute. Turning off the within-layer overlap of a layer’s CPU misses with its GPU hits changes speed by under 1% at every budget, and predicting the next layer’s experts (Hwang et al., 2024; Zhu et al., 2025) never lowers host bytes.

Table 14. The hit-optimal oracle in the engine (job 093; RTX 5090, Ryzen 9 9950X, host memory 50.5 GB/s at best). tok/s of the online policy and of the oracle with the whole sequence in view; their ratio, paired by sequence, with its 95% interval; hit rate and misses per token of each (experts read from host memory by the CPU or by a demand copy), the optimum’s reads in parentheses, and the oracle’s background admissions (Belady prefetch without bypass: it admits what the optimum bypasses); speed as a percentage of this host’s limit; the accounting’s foresight-only term $v(F)$ for the cell in tok/s; the share of that term the oracle recovered and the share of the gap to the limit it closed.

Cell	online	oracle	ratio	hits % (opt.)	misses/token (opt.)	oracle adm.	% of limit	$v(F)$	recovered %	closed %
gpt-oss 11%	47.4	43.7	0.92 [0.91, 0.93]	58 → 67 (73)	60 → 47 (38)	35	47.6 → 44.0	68.8	−27	−16
gpt-oss 25%	77.6	108.5	1.40 [1.38, 1.41]	79 → 92 (89)	30 → 11 (15)	20	31.2 → 43.7	124.4	76	41
gpt-oss 40%	112.4	166.9	1.48 [1.46, 1.51]	89 → 98 (95)	15 → 3 (7)	12	22.1 → 32.8	174.0	92	42
Qwen3 12.5%	27.1	24.8	0.92 [0.91, 0.92]	60 → 66 (75)	155 → 132 (97)	63	48.9 → 44.8	40.2	−28	−18
Qwen3 25%	44.0	54.9	1.25 [1.24, 1.26]	78 → 89 (89)	85 → 42 (43)	46	35.7 → 44.5	71.2	52	31
Qwen3 43.75%	78.7	123.1	1.56 [1.53, 1.59]	92 → 99 (96)	32 → 5 (14)	25	26.1 → 40.8	121.3	103	49

Table 15. The bytes-optimal oracle (MIN with bypass) beside the hit-optimal one on the second oracle host (RTX 5090, Ryzen 9 7950X, host memory 50.6 GB/s at best). Per cell: the online policy’s speed; each oracle’s ratio to it with the whole sequence in view (paired, 95% interval), its hit rate, its misses per token (experts read from host memory by the CPU or by a demand copy; its background admissions in parentheses: the bypass oracle’s are this step’s CPU-served misses, each read a second time by the copy, the hit-optimal one’s are prefetches of what the optimum would bypass as readily as of what it would admit) and the calibrated model on its own counters against its measured time; the optimum’s reads and hit rate from the trace; and MIN with bypass simulated on the trace with the engine’s lookahead (within the sequence) under a copy latency of 1 (ideal), 2 and 3 steps, with the latency at which the simulation meets the engine’s misses.

Cell	online	ratio	bypass oracle, $W=all$				hit-optimal oracle, $W=all$				optimum		simulated misses/token				
			hits	misses (adm.)	model	ratio	hits	misses (adm.)	model	reads	hits	$d=1$	2	3	∞		
gpt-oss 11%	48.0	1.03 [1.03, 1.04]	65	50 (16)	+7%	0.97 [0.96, 0.98]	71	42 (39)	+19%	38.3	73	39	43	47	3.9		
gpt-oss 25%	78.6	1.18 [1.18, 1.19]	86	20 (9)	+6%	1.63 [1.61, 1.66]	95	7 (23)	+53%	15.3	89	16	17	19	4.7		
gpt-oss 40%	113.8	1.23 [1.22, 1.24]	94	9 (4)	+2%	1.56 [1.54, 1.59]	98	3 (12)	+39%	6.8	95	7	8	9	3.5		
Qwen3 12.5%	27.3	1.00 [1.00, 1.01]	64	138 (30)	+5%	0.95 [0.94, 0.95]	67	128 (65)	+11%	96.5	75	97	122	143	2.8		
Qwen3 25%	45.0	1.13 [1.12, 1.14]	83	64 (21)	+9%	1.41 [1.39, 1.43]	91	34 (52)	+37%	43.4	89	44	54	61	3.4		
Qwen3 43.75%	83.2	1.19 [1.18, 1.20]	95	19 (9)	+1%	1.62 [1.58, 1.65]	99	3 (27)	+47%	13.7	96	14	17	19	3.0		

I THE TRACE STUDY IN DETAIL

Policies. Table 16 scores LRU, LFU, decayed frequency, ARC, S3-FIFO and a profiled static cache against MIN and against Belady within a window of W future tokens, at three budgets per model, on each model’s own sampled text (9 traces, the cache carried across conversations). Without admitting every miss, the online policies are close to one another and far from MIN: the best reads 35–110% more, decayed frequency is best at 12 of 27 budget points and S3-FIFO at 10. Where the cache turns over every token ($C/k \leq 1.5$) the six are within 4% of one another; from $C/k=2$ LRU reads 2–11% more than decayed frequency on eight of nine models, and classic LFU fails at large budgets. Adding *admit every miss* (every miss copied into the slot of the lowest-scored resident not requested now) changes the ranking: it is best at 18 of 27 points, ties at 5, and loses by at most 2.7%; it saves up to 5.8% over the previous best, and the best online policy still reads 35–110% more than MIN (prereg/policy_aa.json). At the small budgets almost every resident turns over within a few tokens, and the expert just requested is a good bet. In our engine admitting every miss reads all of its misses over the link, so its value in time depends on the link (Section 5). A static cache profiled on other text reads $1.4\text{--}13.2\times$ MIN because the hot set moves: the C most requested experts of a layer change by 22–58% between windows of 100 tokens and 33–75% between windows of 1,000.

The horizon, before and after the stronger baseline. Against decayed frequency with each admission fetched once (the study’s original baseline), MIN reads 26–56% fewer experts (median 41%), and the window that closes half of the gap fitted $W_{50} \approx 0.59 (C/k)^{1.33}$; admitting every miss lowers the baseline at 23 of 26 points by up to 6.7%, raises W_{50} by a median 9% (at most 42%), and refits it to $0.67 (C/k)^{1.31}$, with a leave-one-model-out exponent range of 1.25–1.36 and, without the points below one token, 1.35 (it was 1.51 against the original baseline). Ninety percent of the gap needs a median of $2.7 C/k$ tokens. Table 17 compares the rule in distinct experts with the power law in tokens and with $W_{50} \propto C/k$, each fitted on eight models and tested on the ninth: the first two are not distinguished, and both beat the third. The rule in distinct experts inverts the held-out model’s own curve of distinct experts against window length, which needs that model’s routing trace.

A learned admission order. We trained a logistic model of whether an expert is requested within the next C/k tokens on features the engine has when it plans a step (decayed request counts at five half-lives, the time since the last request,

Table 16. Host reads per token relative to the exact optimum (Belady’s MIN with bypass over the whole future), median over nine models at three budgets, and the range over all 27 model–budget cells. Event-atomic replay of each model’s own sampled text; every online policy is scored on the last 90% of the trace. Mixtral (8 experts) uses $C = 2, 3$ and 4.

Policy	$C=E/8$	$E/4$	$3E/8$	range
LRU	1.44	1.77	2.03	1.37–2.26
LFU	1.44	1.93	2.40	1.37–6.15
Decayed frequency	1.45	1.70	1.94	1.35–2.10
with hysteresis κ	1.46	1.75	2.00	1.36–2.55
ARC	1.44	1.76	1.99	1.37–2.23
S3-FIFO	1.44	1.69	1.94	1.38–2.11
Static, profiled	1.92	2.74	3.48	1.43–13.24
Belady, $W=1$	1.18	1.45	1.69	1.07–2.06
Belady, $W=4$	1.01	1.13	1.31	1.00–1.95
Belady, $W=16$	1.00	1.00	1.01	1.00–1.66

Table 17. Predicting a held-out model’s horizon W_{50} (fitted on the other eight models, tested on the ninth, over all 26 budget points): error as a factor, $\max(\hat{W}/W, W/\hat{W})$.

Rule	Constants	Median	90th pct.	Max
$W_{50} \propto C/k$	1	1.22	1.57	2.42
$W_{50} = a (C/k)^b$	2	1.19	1.51	1.67
$D(W_{50}) = cC$	1, and $D(W)$	1.16	1.42	1.61

whether the expert is requested now, and transition scores from the previous token’s experts at the same layer and, through an 8-bit table, at every layer), fitted on each model’s mixed-domain trace and replayed on the AIME routing, so no test text is seen in training (Table 18). With each admission read once it reads 6–12% fewer experts than decayed frequency with the same single-read admission at the four host-bound budgets, closing 17–30% of that policy’s excess over MIN; admitting every miss alone closes most of that at Qwen3 12.5% and little at gpt-oss (prereg/learned.offline.json and the research checks). In the engine (job 097, predictions committed before launch) it reads 6–13% fewer experts than the single-read online policy at the host-bound budgets on both hosts, but runs 1.01–1.03 $\times$ it on O4’s machine and 0.90–0.94 $\times$ on host O6: the reads it saves are CPU reads, bought with more fetches in the step, each a copy its layer waits on, and with 237–585 μ s of host time per token (over all six budgets).

Speculation is not foresight. Speculative decoding verifies K draft tokens in one forward pass, so a layer’s experts for K positions are requested together and a missed expert is read once for all of them. On the same traces, with drafts accepted at rate α until the first rejection, batched verification saves an online cache a median 15–32% of its reads at $K=8$ when rejected drafts cost nothing, the equivalent of a window of two to four tokens; once the rejected drafts’ experts are counted, the saving is 8–12% at $\alpha=0.9$ and becomes a loss of 18–20% at $\alpha=0.8$, because the union of K positions’ expert sets is 20–79% of Kk . MIN itself gains little from batching (–7 to 15%). Systems that use drafts as a look-ahead sensor (Li et al., 2026) compare against a baseline that already pays for verification; ours charges the rejected drafts to a cache that does not

Table 18. A learned admission order, offline: host reads per token on the AIME-25 routing the engine runs, of the decayed frequency with each admission read once (single read; within 10% of the engine’s fetch-on-admit reads on hosts O4 and O5), the learned order with single-read admission (trained on the model’s mixed-domain trace; 8-bit cross-layer table, as in the engine), Belady within one and four tokens of perfect foresight, and MIN with bypass in its no-evict form (Table 28). *closed*: the share of the single-read policy’s excess over MIN that the learned order removes. [†]host-bound.

cell	single read	learned	$W=1$	$W=4$	MIN	closed
gpt-oss 11% [†]	59.4	56.0	54.4	47.3	39.0	17%
gpt-oss 25% [†]	28.3	26.1	27.3	25.1	15.5	17%
gpt-oss 40%	14.3	12.6	14.0	13.2	6.8	23%
Qwen3 12.5% [†]	166.2	146.3	134.1	109.1	100.4	30%
Qwen3 25% [†]	84.2	74.8	74.1	62.8	44.3	24%
Qwen3 43.75%	28.8	26.3	27.7	25.3	13.8	17%

Table 19. Partial foresight on the AIME routing the engine runs: host reads per token of admitting every miss and of MIN with one read, and the share of the gap between them that each window closes (Table 2). Recall 0.5: each future expert kept with probability 0.5 and replaced by a random one (fill) or left out (drop); means of three seeds. Linear: the ridge forecaster of Appendix J with a 4-token window. $\kappa=1$: the same exact 4-token window with the deployed margin instead of admitting every miss beyond it, as a share of its own gap.

Budget	Reads per token		Exact window W (tokens)					Recall 0.5, fill / drop		Linear	$\kappa=1$
	every miss	MIN	1	2	4	8	16	$W=8$	all	$W=4$	$W=4$
gpt-oss 11%	58.6	39.3	0.18	0.31	0.52	0.81	0.98	0.29 / 0.29	0.43 / 0.48	0.13	0.62
gpt-oss 25%	28.2	16.0	0.07	0.14	0.25	0.42	0.66	0.14 / 0.13	0.46 / 0.47	0.05	0.34
Qwen3 12.5%	150.0	100.3	0.31	0.51	0.77	0.96	0.99	0.42 / 0.43	0.48 / 0.52	0.19	0.88
Qwen3 25%	79.6	44.9	0.16	0.28	0.46	0.70	0.93	0.29 / 0.28	0.49 / 0.53	0.11	0.60

speculate.

J PARTIAL FORESIGHT IN DETAIL

The window policies. At step g the engine plans the step’s misses before it runs, from the trace’s record of step g (the routers give it layer by layer in a deployable system) and from a view of records $g+1..g+W$ of the same problem. A miss is processed in request order: it takes a free slot; otherwise the victim is a resident not requested now that is not seen in the window (lowest decayed score), or, if every resident is seen, the one seen furthest; a miss seen in the window is admitted unless the victim is seen sooner, and a miss not seen is admitted if the victim is not seen either (admit every miss beyond the window; with the deployed margin $\kappa=1$, only if its score beats the victim’s by one). On the panel admissions are fetched in the step, so every miss is read once; on the deployed path of the follow-up the host’s fetch table splits the misses between the CPU and an in-step copy, and an admission served by the CPU is copied afterwards, a second read. In a degraded view each expert of a future record is kept with probability r , decided by a hash of (seed, g , record, layer, position), so step g sees one view throughout and step $g+1$ an independent one; a dropped expert is replaced by a uniformly drawn one (fill: precision and recall r) or left out (drop: recall r , precision 1). `scripts/value_map.py` replays the plan bit for bit: on a CPU build the engine’s reads equal the replay’s in 30 configurations at three budgets (`cputest099`), and on the panel’s GPUs, where the routing in the run can differ from the recorded trace by numerics, within 1.7% at both gpt-oss budgets. The panel ran the first 20 of the 30 problems; the table’s reads are on all 30. Table 20 lists the windows on the deployed path and in the step on the hosts of job 100. Copied in the step, a window beats the deployed path where the link reads at least 0.92 of the CPU rate and loses to it at 0.54 or below; on the deployed path the engine misses 7–33% more than an instant replay (the deployed state matches its replay within 2.4%), and background copies that land late are the likely cause.

What the table says. Horizon and accuracy trade: at gpt-oss 11% an 8-token window at recall 0.5 closes 0.29 of the gap, about what two exact tokens close (0.31), and the whole future at recall 0.5 closes 0.43. Fill and drop cost about the same for windows of up to 16 tokens; with the whole future in view at low recall, dropping costs less (the largest difference, 0.12, is there). A larger cache needs a longer window: at 25% the same exact 4-token window closes 0.25 of the gap, against 0.52 at 11%.

Table 20. Foresight on each read path (job 100): speed relative to the deployed cache on the same host. *In step*: admit every miss with a window of W exact tokens, each admission copied in the step (as on the panel). *Deployed path*: the deployed policy with the same window, its misses split between the CPU and an in-step copy by the host’s fetch table and its admissions copied in the background; $W=8$ at recall 0.5 in the column marked r . Hosts sorted by the probe’s link-to-CPU ratio.

Host	Link/ CPU	Budget	In step $W=4$	16	Deployed path 4	16	r	MIN, 2 reads	MIN, 1 read	MIN, pref.
Pd again	0.41	gpt-oss 11%	0.66	0.89	1.02	1.03	0.94	1.00	1.05	1.11
Pd again	0.41	gpt-oss 25%	0.69	0.81	1.03	1.12	1.01	1.19	1.11	1.29
9950X, slower link	0.54	gpt-oss 11%	0.65	0.89	1.01	1.01	0.96	0.99	1.07	1.23
9950X, slower link	0.54	gpt-oss 25%	0.69	0.80	0.96	1.06	1.00	1.10	1.11	1.37
5700X3D	0.92	gpt-oss 11%	1.09	1.32	0.99	1.00	0.95	0.99	1.37	1.44
5700X3D	0.92	gpt-oss 25%	1.06	1.23	1.03	1.07	1.00	1.16	1.48	1.80
Ph again	0.96	gpt-oss 11%	0.99	1.20	1.01	1.03	0.98	1.03	1.27	1.45
Ph again	0.96	gpt-oss 25%	0.97	1.09	1.03	1.07	1.01	1.16	1.29	1.64

A realisable forecaster. Ridge regression from the all-layer routing of step t to that of step $t+h$ ($h=1..8$), fitted on each model’s mixed-domain own text and applied to the AIME routing, forecasts the top k experts of each layer with a precision of 0.48 one token ahead and 0.26 four ahead on gpt-oss (0.54 and 0.32 on Qwen3); a decayed request count, the online policy’s own signal, scores 0.30 and 0.28. Driving the window policy, the ridge forecasts close 3–15% of the gap on gpt-oss and 6–19% on Qwen3 over windows of 1–8 tokens (`prereg/forecaster_linear.json`). Its precision falls with the horizon, so over a window it is closer to 0.3 than to its one-token value; at that average precision a window with random errors closes about as much on gpt-oss, and more on Qwen3, where the forecaster’s correct forecasts are mostly experts the cache already holds. A sequence model reports recall above 90% one token ahead with $k+3$ candidates (Wang et al., 2026b); its precision is then at most $k/(k+3)$, 0.57 for gpt-oss and 0.73 for Qwen3. A recurrent forecaster in that spirit (a GRU of 512 units over the same all-layer vectors, one output head per horizon, trained on the same text; `scripts/forecaster_gru.py`) reaches a precision of 0.40 and 0.26 one and four tokens ahead on gpt-oss (0.46 and 0.31 on Qwen3), a recall at $k+3$ of 0.53 and 0.54, and closes 2–11% and 6–17% of the gap: no better than the linear model. Our training text is smaller than SeqMoE’s and differs from the test text, so this bounds what such a model does across text, not what it can do in domain.

K A CLOSED-FORM ACCOUNT OF THE TIME, AND ITS REGISTERED TESTS

This appendix holds the account of the deployed cache’s time that the main text summarises in one paragraph (Section 4): what it is, how it was found, how it fared in the tests registered on machines new to it, and the split of the time it implies. The measured decomposition of Section 4 does not rely on it.

K.1 The relation

For the deployed cache, we describe the time per token T by

$$T = G + \frac{(M + (1 - G/T) A) S}{B_{\text{host}}}, \quad (6)$$

solved for T : it is the larger root of $T^2 - (G + t_M + t_A) T + t_A G = 0$, with $t_M = MS/B_{\text{host}}$ and $t_A = AS/B_{\text{host}}$. M and A are the misses and the background admissions per token (the engine’s counters), S the expert’s bytes, B_{host} the machine’s best probed rate, and G the GPU’s own kernel time per token in an Nsight profile of the same engine (4.1–4.7 ms for gpt-oss; where a machine was not profiled or its profile captured no decode kernels, the median of a fixed set of 7 profiles; Appendix L). G exceeds T_{GPU} of Eq. (2) because it also counts the expert kernels the GPU runs for resident and fetched experts. No constant is fitted to the times. Every host read costs host time, except the background copies that run while the GPU computes: they spread over the next step, and a share G/T of them falls under the GPU’s compute. The *plain* form drops that share. We found the plain form on 30 launches of gpt-oss (exploratory): it is within 6% on 28 of them at 11% but

Table 21. The four registered tests of Eq. (6). A *cell* is one model at one budget on one machine. *New*: machines no earlier launch had rented. Each test registered a band for every cell and a median error of at most 4%; job 108 also allowed at most one cell beyond 6%. Jobs 107 and 108 count only the machines that passed their registered validity checks; job 107 registered that with fewer than three it is inconclusive. *New machines'*: their cells within the band. The last column sets aside, after the fact, the hosts the text discusses: in job 106 the two that computed wrong outputs or ran unsteadily, in job 108 the one whose GPU compute could not be profiled.

Job	Form	Models	Band	Machines (new)	Cells within	New machines'	Median	As registered	Set aside
105	plain	gpt-oss	6%	4 (2)	5/8	2/4	4.2%	failed	–
106	overlap	both	6%	5 (2)	12/20	0/8	4.7%	failed	held (12/12, 2.3%)
107	overlap	both	6%	2 (2)	1/8	1/8	31.7%	inconclusive	–
108	overlap	gpt-oss	8%	3 (3)	4/6	4/6	6.7%	failed	failed (4/4, 5.3%)

over-predicts at 25%, where admissions are a larger share of the reads. The overlap term came from where those copies run, after the plain form failed its first registered test; over 34 launches it brings 33 instead of 18 within 6% at 25%, at the cost of 29 instead of 32 at 11%.

K.2 Registered tests

After the plain form failed, we registered Eq. (6) before three more experiments, each predicting it within a band at every cell, with G profiled on each machine before its timed runs (Table 21). None passed as registered.

The first test: held on machines rented before, failed on new ones. On new runs of 3 machines we had rented before, it came within 6% at all 12 cells of both models (median error 2.3%). This does not single out the overlap term: discounting admissions by half passes 11 of those cells. The same test failed on the two server machines new to it, by 10–52%; one computed wrong outputs and the other ran its rounds 52% apart (slowest over fastest), so the following tests set validity gates before launch (Appendix L).

Machines never rented before: not passed. The next test admitted only machines whose GPU no earlier launch had recorded, with little host memory in use, correct outputs and rounds within 2% of each other. Only two of 7 passed, short of the three required, so by its own rule the test is inconclusive; the relation missed the band at 7 of their 8 cells, by up to 59% on a dual-socket server (51% against the probe's second-highest reading) and 11% on a Ryzen 9 5900XT. A last test restricted the machines in advance to the class the relation had been found on (one NUMA node, at most 32 usable cores) and widened the band to 8%. It failed on an EPYC 7543 inside that class, by 34.3%, and without that machine still failed two pooled clauses (Appendix L).

The error depends on the budget. On the three consumer machines new to the last two tests the relation under-predicted every gpt-oss cell. Over all 25 consumer machines its mean error is -2.2% at gpt-oss 11% (95% interval over machines -3.8 to -0.4) and $+1.6\%$ at 25% (0.2 to 3.1): what Eq. (6) leaves out changes sign with the budget.

K.3 After the fact: the processor

The processor separates most of the machines the relation fits from those it misses, not all; we drew this split after the tests. On 39 valid launches on 25 machines with consumer processors, the read rate the deployed cache implies, $(M + (1 - G/T)A)S/(T - G)$, is 0.88–1.16 of the probe's best rate at gpt-oss 11%, and the relation is within 8% on 37 of them (G was profiled on the machine for 9 of these launches). On server processors, 5 of 8 launches failed a validity check, and the three valid ones read at AMD engineering sample 0.27, EPYC 7543 0.58, EPYC 7402P 1.04 of it: the last fits the relation, the other two do not. We have not established why. On the dual-socket server the probe's own threads read slower with 128 threads, about the engine's 126 helpers, than with 16 (118 against 151 GB/s), and on an EPYC 9655 rented earlier the per-layer timers showed a latency floor no consumer machine has (Appendix D).

K.4 The split it implies (exploratory)

On consumer processors, then, the relation splits the deployed cache's time into three parts (Fig. 10). At gpt-oss 11% the GPU's compute takes 12–45% of the time, MIN's reads, which are the bound, 31–54%, and the reads beyond MIN's 22–35%. The first is serialisation, the GPU's compute in series with the reads, and makes up 23–66% of the gap to Eq. (1);

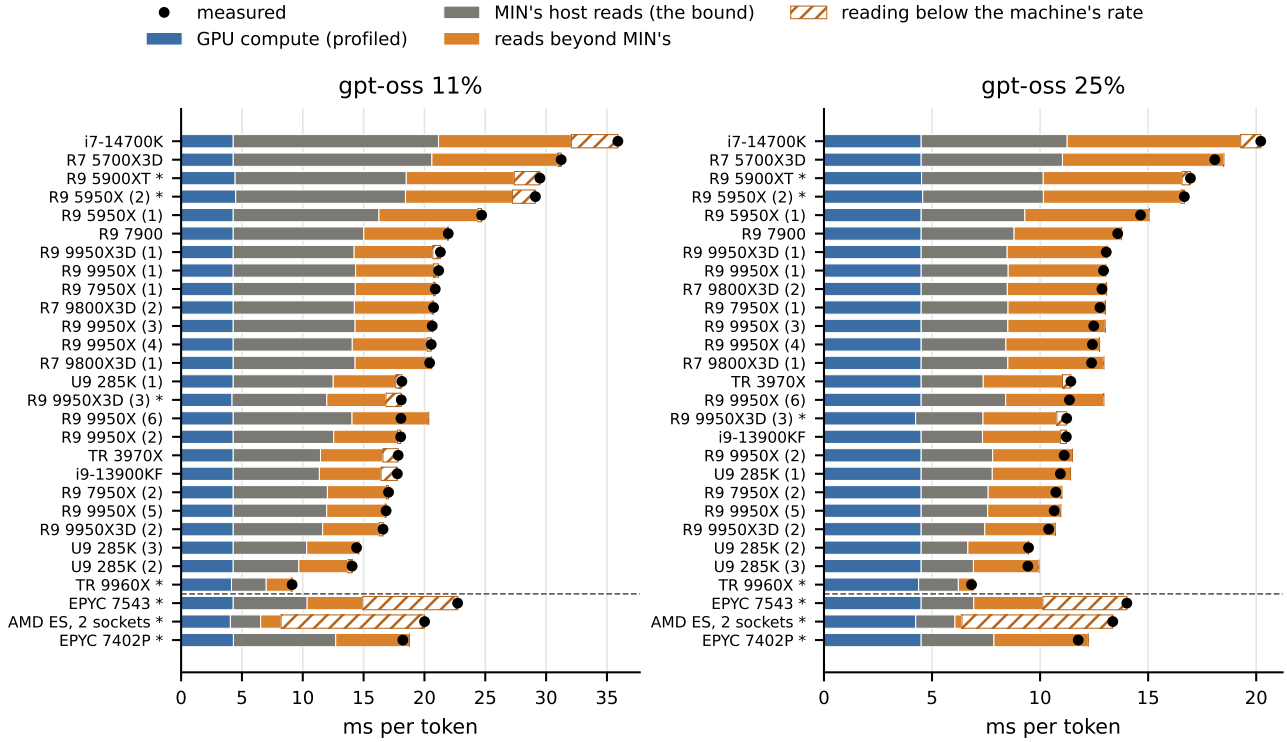


Figure 10. Where the seconds go, machine by machine: the deployed cache’s time per token at the two host-bound gpt-oss budgets split by Eq. (6) into the GPU’s compute (G , profiled on the machine, or the median profile where none was taken or it captured no decode kernels), MIN’s host reads at the machine’s best rate (the bound of Eq. (1)) and the reads the cache makes beyond MIN’s. The dot is the measured time; hatched, the measured time beyond Eq. (6), reading below the machine’s best rate; a bar that ends past the dot is an over-prediction. Machines with the same processor are numbered in order of rental. One launch per machine that ran stably: 25 with consumer processors (above the dashed line) and 3 with server processors (below). *: first rented for a registered test of Eq. (6).

the second is the reads MIN would not make. With G removed, the time scales with the host’s read rate with an elasticity of -0.95 (95% interval -1.06 to -0.84 over the machines of the exploratory set), as reads at B_{host} would. Foresight can attack both parts: fewer reads need MIN’s choices, and overlap needs reads issued ahead. Reading below the probed rate takes at most 11% of the time on the consumer machines; on the server machines it is a third part (EPYC 7402P 0%; EPYC 7543 34%; AMD ES, 2 sockets 59% of the time at 11%).

L MIN’S SCHEDULES AND THE MACHINE, IN DETAIL

The fewest-admission schedule, host by host. Tables 22 and 23 list jobs 104 and 105: MIN’s set loaded by the CPU or in the step, as the greedy rule picks it or as the schedule with the fewest admissions does. On the engine’s 20 problems the fewest-admission schedule copies 0.60–0.73 times as many experts in the step as the greedy one, for 2.5–5.3% more misses (more than its replay predicts; its measured gains include that cost). On every launch it gains over the deployed cache with either load and at both budgets, each 95% interval above 1; loaded by the CPU on the machine with the fastest link, it is 0.07 behind the greedy set at 25%.

Job 106, every host. Table 24 adds the two server machines new to job 106. On the EPYC 7302 host the engine computed wrong outputs: its teacher-forced loss on the deployed cache was 0.21–0.45 nats per token across rounds and models, against 0.19 (gpt-oss) and 0.085 (Qwen3) on every other host and in its own untimed routing pass, and its counters changed between rounds. On the Xeon 8347C host the deployed cache ran its gpt-oss 11% rounds 52% apart (slowest over fastest), against at most 2% registered, and the greedy copy gained in two of its three rounds where we had predicted it would lose. When their jobs reached the gate, before any download, *free* reported 131–132 GB of host memory in use (12–15 GB on the

Table 22. Which of MIN’s hit-optimal sets the oracle follows (job 104): speed relative to the deployed cache on the same host, with MIN’s set loaded by the CPU (two reads) or in the step (one read), as the greedy rule picks it or as the schedule with the fewest admissions does; and the copies per token each makes in the step. Hosts sorted by the probe’s link-to-CPU ratio.

Host	Link/ CPU Budget		By the CPU greedy fewest		In the step greedy fewest		Copies/token greedy fewest	
Pf again	0.29	gpt-oss 11%	0.94	1.11	0.86	1.04	19.8	11.8
Pf again	0.29	gpt-oss 25%	1.13	1.22	0.96	1.06	9.7	7.0
285K, second	0.49	gpt-oss 11%	1.03	1.08	1.05	1.18	19.8	11.8
285K, second	0.49	gpt-oss 25%	1.20	1.21	1.12	1.17	9.7	7.0
Pg again (5950X)	0.63	gpt-oss 11%	1.02	1.06	1.26	1.34	19.8	11.8
Pg again (5950X)	0.63	gpt-oss 25%	1.20	1.15	1.32	1.35	9.7	7.0

Table 23. Job 105, registered before the runs, nothing fitted: the deployed cache’s time per token against the plain form of Eq. (6) (without the overlap term, which came after this job), the GPU’s compute profiled on the same host before the timed runs plus the run’s counted host reads at the machine’s best rate; and the speed, relative to the deployed cache, of the deployed cache with the engine’s layer-ahead copy (each layer’s top predicted expert of the next layer copied ahead of use) and of MIN’s greedy and fewest-admission sets copied in the step. Hosts sorted by the probe’s link-to-CPU ratio.

Host	Link/ CPU Budget		G_{prof} (ms)	Deployed (ms) plain Eq. 6 measured		Layer- ahead	MIN, in step greedy fewest	
Pf again	0.28	gpt-oss 11%	4.1	13.6	13.9	0.72	0.86	1.03
Pf again	0.28	gpt-oss 25%	4.4	9.8	9.5	0.69	0.97	1.06
Threadripper 9960X	0.32	gpt-oss 11%	4.1	9.3	9.1	0.91	0.91	1.04
Threadripper 9960X	0.32	gpt-oss 25%	4.4	7.3	6.8	0.87	0.98	1.04
EPYC 7402P	0.45	gpt-oss 11%	4.3	19.1	18.2	0.81	1.01	1.15
EPYC 7402P	0.45	gpt-oss 25%	4.5	12.9	11.8	0.78	1.09	1.15
O4 again (9950X)	1.04	gpt-oss 11%	4.4	20.8	20.5	1.04	1.36	1.34
O4 again (9950X)	1.04	gpt-oss 25%	4.4	13.2	12.4	1.01	1.42	1.39

other hosts), and their probes’ CPU read rates fell when more threads read than the container had usable physical cores. On both, Eq. (6) under-predicted the deployed cache by 10–52%. We report their clauses (in the supplement) and draw nothing from them about the relation on those runs; both are servers, and job 107’s one valid server was under-predicted as well (Appendix K). The three machines that ran stably were all rented before: Pf, the Threadripper 9960X of job 105, and panel host Pe (same GPU).

Job 107, every host. Table 25 lists the four machines that passed the gate before any download. The EPYC 9754 and the EPYC 7663 ran their gpt-oss 11% rounds 3.5% and 6.0% apart, beyond the registered 2%, and stopped there, as registered; the relation under-predicted them by 31% and 21%. The others stopped at the gate with 50–96 GB of host memory in use. The registration replaced only machines that failed the gate, from a list of offers in order; replacing machines that failed the round check was added during the job, after the results of the first four machines were in, in an amendment committed before the machine it added started. By then one listed offer was gone and the next would have exceeded the remaining credit, so the last two machines were chosen by rules committed before each started (the cheapest verified offer not rented before with cheap traffic and downloads of at least 1000 and 900 Mb/s), and both stopped at the gate. On the two valid machines, the relation predicted the admission margin’s speed from its own counters within a median 0.016 (up to 0.067, on the server), against 0.021 for predicting no change. The profiled GPU compute was 4.1–4.5 ms on gpt-oss and 5.6–5.9 ms on Qwen3, within the range of earlier hosts.

Table 24. Job 106, registered before the runs, every host (*: deployed cache varied by more than the registered 2% between rounds): the deployed cache’s time per token against Eq. (6) (GPU compute profiled on the same host before the timed runs; the run’s own counters), and the speed relative to the deployed cache of the two online admission rules (*dk*: the decayed count with a larger admission margin; *lrn*: the learned reuse predictor with a margin; both chosen on other text) and of MIN’s fewest-admission set copied in the step, with round-level 95% intervals (rounds, then problems: three rounds at gpt-oss 11%, two at 25%; Qwen3 ran one round, so its intervals are over problems). Hosts sorted by the probe’s link-to-CPU ratio.

Host	Link/ CPU Budget		G Deployed (ms)						MIN, fewest in the step	
			(ms)	Eq. 6	meas.	dk		lrn		
EPYC 7302*	0.14	gpt-oss 11%	4.3	12.2	17.7	1.06	[1.01, 1.10]	0.97	[0.92, 1.01]	0.85 [0.82, 0.88]
EPYC 7302*	0.14	gpt-oss 25%	4.7	8.7	12.2	1.08	[1.00, 1.17]	1.10	[1.06, 1.14]	–
EPYC 7302*	0.14	Qwen3 12.5%	5.8	20.4	25.4	1.07	[1.02, 1.13]	0.93	[0.86, 1.00]	–
EPYC 7302*	0.14	Qwen3 25%	6.3	14.3	17.2	1.01	[0.93, 1.08]	0.84	[0.76, 0.94]	–
Xeon 8347C*	0.23	gpt-oss 11%	4.1	12.0	19.5	1.28	[0.99, 1.56]	1.08	[0.80, 1.45]	1.33 [1.00, 1.61]
Xeon 8347C*	0.23	gpt-oss 25%	4.5	8.6	10.1	1.06	[1.05, 1.08]	1.02	[0.99, 1.04]	–
Xeon 8347C*	0.23	Qwen3 12.5%	5.3	21.4	45.0	1.96	[1.67, 2.24]	1.29	[0.94, 1.74]	–
Xeon 8347C*	0.23	Qwen3 25%	5.8	14.9	16.6	0.86	[0.71, 1.02]	0.73	[0.59, 0.88]	–
Pf again (285K)	0.29	gpt-oss 11%	4.3	13.8	13.7	1.02	[1.01, 1.04]	1.03	[1.02, 1.03]	1.02 [1.01, 1.03]
Pf again (285K)	0.29	gpt-oss 25%	4.7	9.6	9.5	1.06	[1.04, 1.07]	1.05	[1.04, 1.07]	–
Pf again (285K)	0.29	Qwen3 12.5%	5.4	24.8	24.4	1.02	[1.00, 1.03]	1.02	[1.01, 1.03]	–
Pf again (285K)	0.29	Qwen3 25%	5.9	16.9	16.2	1.04	[1.03, 1.04]	1.03	[1.02, 1.03]	–
Threadripper 9960X again	0.32	gpt-oss 11%	4.1	9.1	9.4	1.00	[0.99, 1.02]	0.98	[0.97, 1.00]	1.06 [1.05, 1.06]
Threadripper 9960X again	0.32	gpt-oss 25%	4.4	6.9	7.0	1.02	[1.00, 1.04]	0.99	[0.97, 1.00]	–
Threadripper 9960X again	0.32	Qwen3 12.5%	5.4	15.1	14.4	1.02	[1.01, 1.02]	0.99	[0.99, 1.00]	–
Threadripper 9960X again	0.32	Qwen3 25%	5.7	11.2	10.5	1.01	[1.01, 1.01]	0.98	[0.98, 0.99]	–
Pe again (9950X)	1.02	gpt-oss 11%	4.5	20.4	20.5	1.01	[1.01, 1.01]	1.01	[1.01, 1.02]	1.34 [1.33, 1.35]
Pe again (9950X)	1.02	gpt-oss 25%	4.5	12.8	12.4	1.01	[1.01, 1.01]	1.01	[1.00, 1.02]	–
Pe again (9950X)	1.02	Qwen3 12.5%	5.8	35.0	36.7	1.01	[1.01, 1.01]	1.02	[1.02, 1.02]	–
Pe again (9950X)	1.02	Qwen3 25%	5.9	22.2	22.5	1.01	[1.01, 1.01]	1.01	[1.01, 1.02]	–

Job 108, every host. Table 26 lists the machines that ran. The EPYC 7K62 ran its two gpt-oss 11% rounds 7.0% apart and stopped, as registered; the relation under-predicted it by 26.8%. The gate stopped a Ryzen 7 5700X3D whose GPU job 100 had rented, and an EPYC 9354 host with 323 GB of memory in use. Machines were rented from the registered list in order until three were valid. On both EPYC hosts the Nsight trace held almost no kernels (driver 570; the Ryzen hosts ran drivers 580 and later), so their G is the median of the 7 profiles of jobs 069c and 105, as everywhere else (4.28 and 4.50 ms). This is a deviation from the registration that does not change the outcome: without the EPYC 7543 the median error over the 4 cells is 5.3% and all are within 8%, but two pooled clauses still fail. The registered offer list said none of its offers was in the rental ledger, but three were: the 5700X3D’s (job 100; the gate turned it away), the 9950X3D’s (rented for a few minutes in job 099 without producing data; its GPU matched no earlier launch’s) and an i9-13900KF’s that was never reached.

The probe’s rate, and machines as the unit. B_{host} is the highest single reading of one probe run. Its second-highest reading is a median 0.8% lower and more than 5% lower on 1 of 42 launches (28 machines): launch 107d (Table 25), by 26% (Table 27). With the second-highest reading, Eq. (6) is within 6% on 34 instead of 31 of the 39 consumer launches at gpt-oss 11%; with the best CPU-only reading, which ignores that the link reads at the same time, on 22. By Eq. (6)’s own accounting the engine reads more than 5% faster than the highest reading on 12 of 84 launch-budgets (up to $1.22\times$), so the bounds are relative to the probe. Taking the machine as the unit, one value per machine: the fewest-admission schedule copied in the step gains a mean $1.21\times$ (95% interval over machines 1.13–1.30) on the 9 machines that ran stably; the admission margin, each machine’s mean over the two gpt-oss budgets, $1.020\times$ (1.011–1.030) on 5. With so few machines the percentile interval is narrow; the t -intervals on the same values are 1.11–1.33 and 1.004–1.035. Equation (6)’s error is a mean -2.2% (-3.8 to -0.4%) at gpt-oss 11% over the 25 consumer machines, and $+1.6\%$ (0.2 to 3.1%) at 25%; on the 21 machines it was found on, -1.5% (-3.2 to 0.4%) at 11%: of the 30 launches it was found on, it under-predicts 25 at 11% and over-predicts 23 at 25%. G was profiled on the machine for 9 of the 39 consumer launches; the others use the median of the profiles. The median |error| is 3.2% on the launches with their own G and 1.7% on the others, but the profiled launches all come from the registered tests.

Table 25. Job 107, registered before the runs, on the machines never rented before that passed the registered gate before any download ($\dagger$: rounds further apart than the registered 2%; such a host stopped after its gpt-oss 11% rounds and is outside the predictions): the deployed cache’s time per token against Eq. (6) (GPU compute profiled on the same host before the timed runs; the run’s own counters), and the speed relative to the deployed cache of the admission margin (dk) and of MIN’s fewest-admission set copied in the step or loaded by the CPU, with round-level 95% intervals (rounds, then problems: two rounds of gpt-oss, one of Qwen3, whose intervals are over problems). Hosts sorted by the probe’s link-to-CPU ratio.

Host	Link/ CPU Budget		G Deployed (ms)				MIN, fewest admissions			
			(ms)	Eq. 6	meas.	dk	in the step		by the CPU	
EPYC 7663 $\dagger$	0.21	gpt-oss 11%	4.3	11.3	14.3	1.03 [1.00, 1.05]	0.97 [0.94, 1.01]	1.13 [1.07, 1.18]		
AMD engineering sample	0.44	gpt-oss 11%	4.1	8.2	20.0	1.05 [1.04, 1.06]	1.16 [1.14, 1.17]	1.13 [1.11, 1.14]		
AMD engineering sample	0.44	gpt-oss 25%	4.2	6.4	13.4	0.99 [0.98, 1.00]	–	–		
AMD engineering sample	0.44	Qwen3 12.5%	5.6	13.6	32.7	1.03 [1.02, 1.04]	–	–		
AMD engineering sample	0.44	Qwen3 25%	5.9	10.4	23.7	1.07 [1.06, 1.09]	–	–		
EPYC 9754 $\dagger$	0.51	gpt-oss 11%	4.2	10.1	14.7	0.97 [0.95, 0.99]	1.18 [1.15, 1.20]	1.05 [1.02, 1.08]		
Ryzen 9 5900XT	0.75	gpt-oss 11%	4.4	27.4	29.5	1.02 [1.02, 1.02]	1.42 [1.41, 1.43]	1.04 [1.04, 1.04]		
Ryzen 9 5900XT	0.75	gpt-oss 25%	4.5	16.6	17.0	1.02 [1.01, 1.02]	–	–		
Ryzen 9 5900XT	0.75	Qwen3 12.5%	5.8	48.3	54.4	1.02 [1.02, 1.02]	–	–		
Ryzen 9 5900XT	0.75	Qwen3 25%	5.9	29.7	31.7	1.02 [1.02, 1.02]	–	–		

Table 26. Job 108, registered before the runs and restricted in advance to desktop-class machines never rented before: the deployed cache’s time per token (gpt-oss, mean of two rounds) against Eq. (6), with G profiled on the same machine before the timed runs (on the two EPYC hosts, whose traces held no decode kernels, the median of the profiles of jobs 069c and 105), the run’s own counters and the probe’s best rate, and the read rate the time implies as a share of that rate. $\dagger$: rounds further apart than the registered 2%; outside the predictions.

Machine	Cores	B_{host}	Budget	G (ms)	Eq. (6)	Measured	Implied/best
EPYC 7K62 $\dagger$	23	22	11%	4.3	46.2	63.1	0.71
Ryzen 9 5950X	16	36	11%	4.5	27.2	29.1	0.92
Ryzen 9 5950X	16	36	25%	4.6	16.5	16.7	0.99
Ryzen 9 9950X3D	16	65	11%	4.2	16.8	18.1	0.91
Ryzen 9 9950X3D	16	65	25%	4.2	10.8	11.2	0.94
EPYC 7543	30	84	11%	4.3	14.9	22.7	0.58
EPYC 7543	30	84	25%	4.5	10.1	14.0	0.61

Admitting less, in detail. The margin κ was chosen as the value that minimises the deployed path’s reads on each model’s mixed-domain text, not the AIME text (`scripts/online_admit.py`): 3 at the smallest budget and 2 at 25%. It cuts the deployed cache’s host reads by 4–6% at gpt-oss 11% and 6–10% at 25%. On the three machines of job 106 that ran stably it ran 1.00–1.02 $\times$ and 1.01–1.06 $\times$ the deployed cache (Qwen3: 1.01–1.04 $\times$; Table 24). We had registered at least 1.01 at gpt-oss on every host: it fell short on two of the three stable ones, and the registered pooled median of at least 1.02 held only through the two unstable machines (1.01 on the stable ones). On the two valid machines of job 107 it ran 0.987–1.074 $\times$ (median 1.021), below the registered 0.995 at one cell, the server at gpt-oss 25%. Predicted from each configuration’s own counters, Eq. (6) gives these ratios within a median 0.013 on job 106’s stable machines, but predicting no change does almost as well (0.016), and on job 107’s server it missed the registered 0.03 at two cells. A learned reuse predictor with the same kind of margin (job 097’s, fitted on other text) reads less still (7–14% fewer) but costs 267–439 μs of host time per token on gpt-oss and runs 0.98–1.05 $\times$: it pays where host memory is slow and loses where it is fast, against the registered prediction that it would not lose.

The slowest link. On the EPYC 7302 host of job 106, whose link moves 13 GB/s, the fewest-admission set copied in the step lost in each round (0.82–0.88 $\times$); that host computed wrong outputs throughout, so we draw nothing from it, but below some link rate even the fewest-admission set should not be copied in the step.

The machine decides how much. Across machines, the gain of every state that moves reads from the CPU to the link follows the link-to-CPU ratio (Spearman over the machines that ran it through job 104, 95% bootstrap interval): MIN copied in the step 0.89 [0.63, 0.97] (19 machines), copied ahead 0.87 [0.54, 0.99] (17), admitting every miss 0.95 [0.75, 0.99] (13). MIN loaded by the CPU, which reads every admission twice and copies it in the background, tracks the link’s absolute

Table 27. How the probe’s rate changes the picture, over the 84 launch-budgets at gpt-oss 11% and 25% of the machines that passed our checks: Eq. (6) with the highest probe reading (as in the paper), the second-highest, and the best CPU-only reading. *Faster than the rate:* launch-budgets whose implied read rate exceeds the rate by more than 5%. The error columns are the consumer launches at 11%.

Rate	Faster than the rate (max)	Within 6%	Within 8%	Median error
highest reading	12 (1.22 $\times$)	31/39	37/39	1.7%
second-highest	15 (1.26 $\times$)	34/39	37/39	1.5%
best CPU-only	46 (1.53 $\times$)	22/39	29/39	5.6%

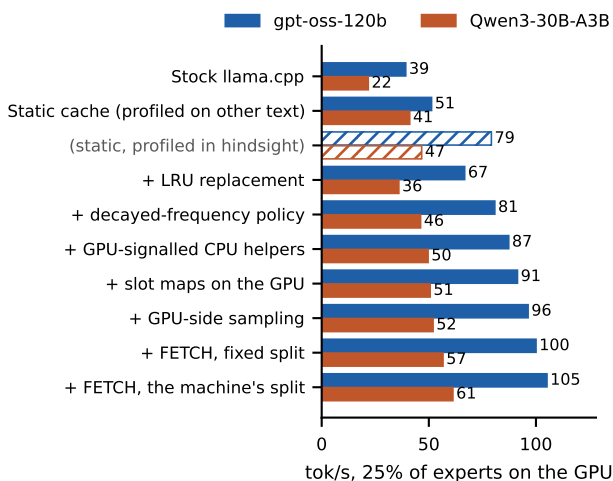


Figure 11. Ablation at 25% of experts on the GPU (15 AIME-25 problems, 256 tokens, host A). Each step adds one component to the one above; the hatched row is not a step: a static cache of each layer’s 32 most requested experts counted on the measured problems themselves.

rate more closely (0.79 [0.49, 0.92], 17 machines, against 0.45 for the ratio; the two intervals overlap, and the feature was picked from a search over all probe features). The greedy schedule copied in the step gains 1.02–1.05 $\times$ at 11% on 2 further machines with ratios 0.40 and 0.48, and loses only on the machines whose link reads at under a third of their CPU rate; made ahead of use, the same copies lose beyond their interval on no machine they ran on (at worst 0.997 $\times$, on Pf).

Machines differ more than problems. Problems rank largely the same on every machine: Kendall’s coefficient of concordance of MIN-copied-in-the-step’s per-problem gain over 19 machines and 20 problems is 0.77 at 11% and 0.82 at 25%, and a problem’s gain follows how far the deployed policy’s reads exceed MIN’s on its trace (Spearman 0.88 at 11%, 0.68 at 25%). Yet machines carry 96% of the variance of its log gain and problems 1–3%.

M THE EXPERT CACHE IN MORE DETAIL

This section holds the measurements of Section 2 that the main text summarises: where the cache’s gain comes from, the choice of the FETCH split, the hosts with half the link, long outputs, FreeToken’s own headline model, two other llama.cpp-based systems, output parity, and the cost of prompts and batches. Job numbers refer to the scripts on the gpu branch.

Where the gain comes from. Figure 11 builds the system up from stock llama.cpp on one machine. Placing each layer’s hot experts is the first gain: a static cache of the 32 experts per layer most requested on other text (the models’ own MATH-500 outputs) runs 1.31 $\times$ and 1.89 $\times$ stock, which pins whole layers instead. Its hit rate (52% and 67%) depends on how well the profile matches the workload: profiled in hindsight on the measured problems it hits 75% and 72% and runs faster (hatched). Replacing experts online closes that gap without a profile, if admissions are rationed: LRU admits every miss, and on Qwen3 its copies raise host reads from 127 to 188 experts per token, so it runs 0.88 $\times$ the static cache

Table 28. The bound under each tightening, tokens per second on host B, and our cache as a percentage of the bound as published (exact optimum, highest probed host rate, datasheet GPU rate) and with every tightening at once. *no-evict*: Belady’s MIN with bypass that keeps a served expert for the step; *exact*: the same, free to evict an expert after serving it; *pool*: one pool of *LC* slots shared across layers; *median*: B_{host} the median of six concurrent probe samples instead of the highest; *GPU*: the GPU term from the measured all-in-VRAM time instead of the datasheet rate; *layers*: the per-layer sum, which no system without cross-layer prefetch can beat; *all*: exact, pool, median and GPU together (a warm start, not included, changes the bound by at most 1.4%).

Model	Experts on GPU	no-evict	exact	pool	median	GPU	layers	all	Ours, % of exact	all
gpt-oss-120b	11%	169	172	182	159	172	163	168	41	41
gpt-oss-120b	25%	427	430	474	397	279	365	278	25	39
gpt-oss-120b	40%	519	519	519	517	279	500	278	29	55
Qwen3-30B-A3B	12.5%	92	96	101	89	96	93	93	42	43
Qwen3-30B-A3B	25%	209	214	234	197	192	193	191	30	33
Qwen3-30B-A3B	43.75%	308	308	308	307	192	298	191	35	57

(1.30 $\times$ on gpt-oss); the decayed-frequency policy reads 41 and 118 per token, runs 1.58 $\times$ and 1.12 $\times$ the profiled static cache, and the hindsight cache runs 0.98 [0.95, 1.01] and 1.01 [0.98, 1.04] times as fast as it. On traces LRU misses only 2–11% more than decayed frequency at these budgets (Appendix I); the rest of its cost here is the copies it admits. On host S, with everything else as in the full system, LRU runs 6–21% slower than decayed frequency across the six cells of Table 4 and trails FreeToken at four of them: the admission policy is most of the lead over FreeToken (77–84% of it at gpt-oss 25 and 40%, more than the whole of it at gpt-oss 11% and Qwen3 12.5 and 25%). The GPU-signalled CPU helpers, GPU slot maps, GPU sampling and FETCH add 2–9% each, and the model’s FETCH table 5–8% over the fixed one. The full system runs 2.68 $\times$ (gpt-oss) and 2.81 $\times$ (Qwen3) stock llama.cpp.

How tight is the bound? Table 28 changes one assumption at a time on host B. Pooled slots need 4.5–18.6% fewer reads and raise the host-bound limits by 5–10%; starting MIN warm changes it by at most 1.4%; the median probe sample instead of the highest lowers the host-bound limits by 8%. The GPU term moves most: at batch size 1 the all-in-VRAM runs reach only 52% (gpt-oss) and 61% (Qwen3) of the datasheet rate, and with the measured rate the limits at 25% and above fall to 279 and 192 tok/s, 8–9% above the all-in-VRAM speed. In the running example (gpt-oss at 25%) the limit on host B is 430 tok/s (279 with the measured GPU rate) and the cache runs 109 tok/s. With every expert in its slots the cache matches stock llama.cpp (0.997 $\times$ and 1.047 $\times$). The bound assumes an expert run on the CPU is read from DRAM (MIN’s shortest re-read distance is 0.5 GB of host traffic) and that computation runs at memory speed. The routing traces are each model’s own greedy text on the 30 prompts; replayed through a simulator of the deployed policy they reproduce the engine’s hit rates within 5.3 points.

Reaching the read time. The microbenchmark of Section 3 (`jobs/ec2/readsched.cu`) takes MIN’s misses per layer and step from `scripts/readsched.py` and runs no model. Usable cores minus two CPU threads read their share of host memory from an ordinary buffer, as the engine’s helpers do, and the copy engine copies the rest, 13.25 MB at a time, from pinned memory into GPU slots; both buffers rotate through 4 GiB so that no cache holds them. *Per layer* splits each layer’s experts, whole, between the two paths at the probe’s rates and waits for both before the next layer; *per token* sums the same split over the token and waits once; *link only* sends every read over PCIe, as admitting every miss does (MIN with one read sends only its admissions); *CPU only* reads everything on the CPU. Table 29 gives each host. Waiting per layer instead of per token costs at most 1 point of the host term on the two AMD hosts and 7 on the Intel host, whose link is the slowest; there, reading everything on the CPU beats the per-layer split. The split ignores which reads are admissions; the link-aware bound sends every admission over the link and raises the bound by at most 1% at 11% (`scripts/linkaware.py`, `prereg/linkaware.json`).

Choosing the split per machine. On host B, the model’s table beats the fixed table we used before at every budget of both models, by 3.3–8.0% (launch 1). Table 30 shows why at one budget: a single missed expert should run on the CPU

Table 29. Can the bound’s read time be reached (job 102)? MIN’s per-layer host reads of gpt-oss (first 4,000 steps of the AIME routing), replayed without the model through CPU threads and the copy engine. Each mode’s share of the bound’s host term (R^*S/B_{host} , ms per token in the column *Term*); *Model*: the per-layer mode predicted from the probe’s rates. Two repetitions of each mode agree within 0.9%.

Host	Link/ CPU	C	Term (ms)	Split per layer	Split per token	Link only	CPU only	Model
Pd again	0.40	14	7.04	86%	93%	35%	92%	98%
Pd again	0.40	32	2.84	89%	93%	35%	92%	98%
9950X	1.02	14	9.70	96%	97%	95%	86%	94%
9950X	1.02	32	3.91	96%	97%	94%	86%	92%
9800X3D	0.55	14	9.56	95%	95%	53%	91%	95%
9800X3D	0.55	32	3.85	93%	94%	53%	91%	94%

Table 30. FETCH tables at 43.75% on Qwen3-30B-A3B (launch 1): entry n is how many of a layer’s n missed experts are copied over PCIe; the rest run on the CPU. Ratio to the fixed table, paired, 95% CI.

Table	$f(0..8)$	tok/s	vs fixed
Lite	0, 0, 1, 1, 1, 2, 2, 3, 3	109.0	1.057 [1.052, 1.062]
Model	0, 0, 1, 1, 2, 3, 3, 4, 5	108.0	1.048 [1.041, 1.055]
Model, fetch a single miss	0, 1, 1, 1, 2, 3, 3, 4, 5	104.1	1.010 [1.007, 1.012]
Fixed	0, 1, 1, 2, 3, 3, 4, 5, 6	103.1	1
None (CPU only)	0, . . . , 0	100.7	0.977 [0.970, 0.984]
All (fetch every miss)	0, 1, . . . , 8	98.7	0.957 [0.953, 0.962]

(about 155 μ s, hidden behind the GPU’s other experts) rather than be copied (about 225 μ s, on the critical path); that entry alone is worth 3.7%. Fetching everything, FreeToken’s offload strategy expressed in our engine, is the worst table, and running everything on the CPU the second worst. The model’s table was within 1% of the best of five alternatives on launch 1 and tied with it on launch 2. Given each run’s measured misses, Eq. (5) ranks the tables correctly except the model’s table against the lite one, which are within 1%, and overstates their differences about 1.8 $\times$.

Machines with half the link. On RTX 5090s whose links we measured at 27–30 GB/s instead of 53 (jobs 085, 085b and 088: gpt-oss next to a Core i9-14900K, Qwen3 next to a Ryzen 9 7900 in a PCIe 4.0 slot), the model’s tables fetch less (gpt-oss 0, 0, 0, 1, 1, Qwen3 0, 0, 1, 1, 2, 2, 3, 3, 4) and beat the fixed table by 21–34% on gpt-oss and 9–18% on Qwen3, about twice the gain on host B for Qwen3 and four to six times for gpt-oss: the per-machine split matters most where the link is slow. On the Ryzen, ours leads FreeToken by 1.04–1.08 $\times$; with the fixed table it would trail at 12.5%. On the Core i9 the first comparison (job 085) was confounded: FreeToken’s hybrid backend ran gpt-oss at 12–66 tok/s, below its own offload backend (25–86), because its CPU executor spread 23 threads across the 8 performance and 16 efficiency cores and its calibration then sent two thirds of the misses over the slow link, the same failure Appendix D reports for llama.cpp; the 1.4–2.0 $\times$ lead that gave us measured a thread configuration. We reran it (job 088, a second rental of the same CPU model) with FreeToken’s executor on 8 threads pinned to the performance cores: its hybrid backend then runs 1.4–3.1 $\times$ faster than before (37.5, 62.1 and 92.5 tok/s; its calibration now copies 51.7% of misses), beating its own offload backend at 11 and 25% and tying it at 40%, and ours leads its better configuration by 1.58, 1.52 and 1.51 $\times$ (lowest interval bound 1.47). Restricted to the CPU alone, FreeToken’s 8-thread executor decodes 13.7 tok/s at every budget. llama.cpp on the same rental runs 40.0 tok/s with one thread per performance core (−9.4% from the model’s prediction) and 32.8 with all 24

threads (+25.9%): the hybrid-core failure is a thread-count choice, and the model holds once the choice is right. The model predicted llama.cpp on the Ryzen within +2.7%. Figure 7 puts every same-machine comparison on one axis.

Long outputs. Our lead does not come from the first 256 tokens. On gpt-oss, on ten held-out hard problems (AIME 2022, problems 11–15 of both exams) that neither system has finished reasoning about after 2,048 tokens (at most 1 of 10 reach an answer), ours runs 1.36 [1.34, 1.38], 1.32 [1.30, 1.34] and 1.19 [1.17, 1.21] times FreeToken at 11, 25 and 40% over all 2,048 tokens, and 1.37 [1.34, 1.39], 1.33 [1.31, 1.35] and 1.19 [1.17, 1.22] over tokens 257–2,048 (job 083, host A); both engines stream one event per token. On Qwen3, over tokens 257–2,048 of ten held-out MATH-500 problems, ours runs 1.043 [1.020, 1.070], 1.056 [1.036, 1.077] and 1.054 [1.025, 1.086] times FreeToken at 12.5, 25 and 43.75%. Both leads are larger than over the same runs’ first 256 tokens. Turning off FreeToken’s prefill overlap, which lends it cache slots during each prompt, changes its overall speed by 1% or less.

FreeToken’s own headline model. On FreeToken’s own headline model, Qwen3.6-35B-A3B in BF16 (256 experts of 6.3 MB per layer, hybrid linear attention), the two systems are close: ours runs 1.030 [1.023, 1.038] times FreeToken at 12.5% and ties at 25 and 37.5% (1.010 [0.998, 1.021], 1.003 [0.994, 1.013]), at 1.98–2.48× llama.cpp (job 086, host A). FreeToken as shipped, with its automatic backend and cache size, runs this model at 93.8 tok/s on that machine, above the 77–83 tok/s its paper reports on an RTX 5090, so our setup does not handicap it; FreeToken’s fetch cap and thread count were left at their defaults, while our FETCH table was swept. Our lead is therefore model-dependent: largest with few large experts (gpt-oss, top-4 of 128), smallest with many small ones.

FreeToken tuned. On an RTX 5090 with a Ryzen 9 9950X (job 098, one launch, configurations in a shuffled order per cell) FreeToken ran in five settings per cell: hybrid with its calibrated fetch split (its default), hybrid with the fetch cap fixed at one and at two, hybrid with 8 CPU threads instead of one per physical core, and offload. Its best setting was the calibrated hybrid with 8 threads at three cells and offload at the other three (gpt-oss 25 and 40%, Qwen3 43.75%), at most 1.04× its Table 4 setting; the fixed caps were never better than its own split by more than 0.3%. Ours over FreeToken’s best: 1.17, 1.17 and 1.09 on gpt-oss, 1.02, 1.04 and 0.98 on Qwen3, within 0.04 of host S’s ratios at every cell. With every expert on the CPU, stock llama.cpp decodes gpt-oss 1.13× and Qwen3 0.99× as fast as FreeToken’s CPU backend.

Other systems and outputs. Two newer llama.cpp-based systems, Pipelined Sharding (Ukarande et al., 2026) and leloch’s expert cache (leloch, 2026a), run gpt-oss-120b at 1.28–1.45× stock llama.cpp; ours ran 1.42–2.34× the better of them (job 077). KTransformers (Chen et al., 2025a) (kt-kernel 0.7.0, exact BF16 CPU experts, the same number of experts statically on the GPU) ran Qwen3 in BF16 at 0.98–1.01× stock llama.cpp on an RTX 5090 with a Ryzen 9 9950X, and ours at 2.2–4.4× it (job 078); both static placements read the same CPU expert bytes from the same DRAM. Its published speeds rely on AMX Xeons and INT4/INT8 CPU weights, which that host lacks and which change the model, so this is not its best setting. Teacher-forced on fixed text (1536 steps per model) against stock llama.cpp with every weight in the VRAM of an RTX PRO 6000 (job 090), our cache at 25% has a mean KL divergence over the full vocabulary of 0.0019 nats on gpt-oss and 0.0005 on Qwen3 (99.9th percentile 0.121 and 0.059, maximum 0.21 and 0.13), picks the same next token on 98.5% and 99.3% of steps, and changes the loss by +0.16% and +0.19%. Stock llama.cpp’s own --n-cpu-moe placement diverges from the same reference by as much (mean KL 0.0017 and 0.0008, agreement 98.8% and 99.3%): the differences are those of running experts on the CPU (the helpers quantise activations to Q8_0 where the GPU path uses Q8_1), not of the cache. Task accuracy was not measured.

Prompts and batches. The cache serves single-token steps; prompts and batches above one run llama.cpp’s stock graph with every expert in host memory, and pay for the cache’s resident experts and its one-thread CPU side. In llama-batched-bench at 25% on host S (job 089, both models) our prefill at 512 and 2,048 tokens runs 79–83% of stock llama.cpp’s and our decode at 2 and 4 parallel sequences 77–79% of it (the tool feeds both engines the same random tokens).

N THE AUDIT’S ROWS

Table 31 lists the 52 published measurements of Section 3 with their bounds. The sources are Fiddler (Kamahori et al., 2024), ProMoE (Song et al., 2024a), KTransformers (Chen et al., 2025a), HybriMoE (Zhong et al., 2025), Fate (Fang et al., 2025), SP-MoE (Chen et al., 2025b), MoE-SpeQ (Wang et al., 2025), the CPU–GPU collaborative inference of Huang et al. (2025), FreeToken (Yang et al., 2026), SeqMoE (Wang et al., 2026b), NVIDIA’s pipelined sharding (Ukarande et al., 2026),

Where the Seconds Go

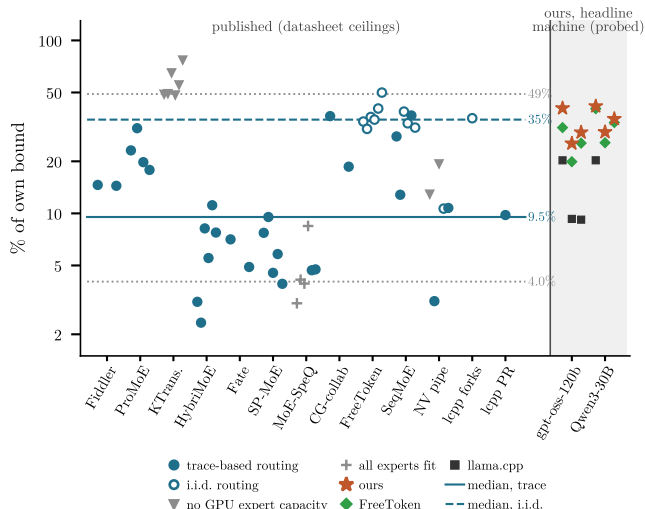


Figure 12. Published MoE-offloading measurements (52 rows from 13 sources) as a percentage of their own bound, computed from the paper’s hardware at datasheet rates and, where a routing trace of the model exists, the per-layer Belady optimum (filled); hollow points assume uniform routing, which overstates the optimum’s reads and so the percentage; the dotted lines are the class medians (no GPU expert capacity, 49.0%; every expert resident, 4.0%). Right: our cache, FreeToken and stock llama.cpp on host B against the probed per-layer limit of Table 4 (the pooled bound FreeToken’s and llama.cpp’s designs allow is 0–10% higher).

a llama.cpp fork (leloch, 2026a) and a llama.cpp pull request (csantiago78, 2026). The in-class subset of Section 3 (20 rows) is the trace-scored rows less the lossy (Fate) and speculative (SP-MoE, MoE-SpeQ) ones; over all 29 trace-scored rows the median is 9.5%. These systems mostly read on demand, so Eq. (2) would place them higher. Three caveats apply to every row: the ceilings are datasheet ones where ours are probed; the trace is the model’s own sampled text, sometimes of a sibling variant (an FP8 trace for a Q4 deployment); and systems whose cache is pooled across layers, whose placement pins whole layers, or which verify several tokens per pass are scored against the per-layer one-token bound, which is not theirs.

Table 31: The audit’s rows (Section 3): every published batch-1 measurement we scored, with the system’s stated hardware, its experts per layer resident on the GPU (C), the model’s experts and top- k , the published speed, the bound at datasheet-physical ceilings (GPU datasheet bandwidth, the top of the host-DRAM band, datasheet PCIe, no latencies; `scripts/audit.py`), how the optimum’s reads were obtained (*trace*: Belady’s MIN with bypass on an on-policy routing trace of the model; *uniform*: independent uniform routing, which overstates the reads and so the percentage; *no slots*: no GPU expert capacity, the CPU-only rate is the bound; *all fit*: every expert resident, nothing offloaded) and the percentage of the bound. Medians: trace 9.5% ($n=29$), uniform 34.8% (11), no slots 49.0% (8), all fit 4.0% (4), all rows 14.5% (52). Rows below 5% are more likely configuration mismatches with the bound’s assumptions (precision, dense layers on the CPU, end-to-end timing) than inefficiency, and are not adjudicated.

System	Model	GPU	C	E/k	tok/s	bound	reads	%
Fiddler (Kamahori et al., 2024)	Mixtral-8x7B	Quadro RTX 6000	1	8/2	2.5	17	trace	14.6
	Mixtral-8x7B	RTX 6000 Ada	3	8/2	8.3	57	trace	14.4
ProMoE (Song et al., 2024a)	DeepSeek-V2-Lite	RTX 4090	45	64/6	49.9	216	trace	23.1
	Qwen1.5-MoE-A2.7B	RTX 4090	46	60/4	67.2	216	trace	31.1
	Qwen2-57B-A14B	RTX 4090	42	64/8	26.9	136	trace	19.8
	Mixtral-8x7B	RTX 4090	6	8/2	27.0	152	trace	17.8
KTransfomers (Chen et al., 2025a)	DeepSeek-V3-0324	A100 40GB	0	256/8	5.9	12	no slots	48.8
	DeepSeek-V2.5-1210	A100 40GB	0	160/6	11.9	24	no slots	49.2
	Qwen2-57B-A14B	A100 40GB	0	64/8	22.9	35	no slots	64.9
	DeepSeek-V3-0324	RTX 4080	0	256/8	16.1	33	no slots	48.3
	DeepSeek-V2.5-1210	RTX 4080	0	160/6	17.6	32	no slots	55.3

continued

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System	Model	GPU	C	E/k	tok/s	bound	reads	%
HybriMoE (Zhong et al., 2025)	Qwen2-57B-A14B	RTX 4080	0	64/8	37.0	48	no slots	76.9
	DeepSeek-V2-Lite-Chat	RTX A6000	16	64/6	14.5	470	trace	3.1
	DeepSeek-V2-Lite-Chat	RTX A6000	48	64/6	15.3	655	trace	2.3
	Mixtral-8x7B-Instruct	RTX A6000	2	8/2	4.4	53	trace	8.2
	Mixtral-8x7B-Instruct	RTX A6000	6	8/2	7.3	133	trace	5.5
Fate (Fang et al., 2025)	Qwen2-57B-A14B-Instruct	RTX A6000	16	64/8	9.6	86	trace	11.1
	Qwen2-57B-A14B-Instruct	RTX A6000	48	64/8	9.0	116	trace	7.7
	Qwen1.5-MoE-A2.7B	RTX 3090	46	60/4	14.0	198	trace	7.1
	Qwen1.5-MoE-A2.7B	GTX 1080 Ti	15	60/4	3.7	76	trace	4.9
	Mixtral-8x7B	RTX 3090	1	8/2	1.2	16	trace	7.7
SP-MoE (Chen et al., 2025b)	Mixtral-8x7B	RTX 4090	1	8/2	1.3	13	trace	9.5
	Mixtral-8x7B	A100 40GB	3	8/2	1.5	32	trace	4.5
	Phi-3.5-MoE	RTX 4090	3	16/2	3.3	57	trace	5.8
	DeepSeek-V2-Lite	A100 40GB	7	64/6	5.6	142	trace	3.9
	DeepSeek-V2-Lite	A100 40GB	64	64/6	10.1	335	all fit	3.0
MoE-SpeQ (Wang et al., 2025)	DeepSeek-V2-Lite	A100 40GB	64	64/6	13.8	335	all fit	4.1
	Qwen1.5-MoE-A2.7B	A100 40GB	60	60/4	13.2	335	all fit	3.9
	Qwen1.5-MoE-A2.7B	A100 40GB	60	60/4	28.3	335	all fit	8.5
	Phi-3.5-MoE	A100 40GB	7	16/2	6.0	129	trace	4.7
	Phi-3.5-MoE	A100 40GB	7	16/2	6.1	129	trace	4.7
CPU-GPU collab. (Huang et al., 2025)	Mixtral-8x7B	RTX 4090	1	8/2	4.8	13	trace	36.5
	Phi-3.5-MoE	RTX 4090	3	16/2	10.4	56	trace	18.6
	Qwen3.6-35B-A3B	RTX 5090	94	256/8	77.1	227	uniform	34.0
	DeepSeek-V4-Flash	RTX 5090	28	256/6	24.9	81	uniform	30.8
	Qwen3.6-35B-A3B	RTX 5090	104	256/8	73.8	205	uniform	36.1
FreeToken (Yang et al., 2026)	Qwen3.6-35B-A3B	RTX 4090	72	256/8	42.9	123	uniform	34.8
	Qwen3.6-35B-A3B	RTX 4060 Laptop	60	256/8	39.3	97	uniform	40.4
	GLM-5.2	RTX PRO 6000 Blackwell	36	256/8	14.9	30	uniform	49.9
	Qwen3-30B-A3B-FP8	RTX 4090	57	128/8	104.1	374	trace	27.9
	Qwen3-30B-A3B-FP8	RTX 4090	19	128/8	47.9	374	trace	12.8
SeqMoE (Wang et al., 2026b)	Qwen3.6-35B-A3B-BF16	RTX 5090	102	256/8	118.3	305	uniform	38.8
	Qwen3.6-35B-A3B-BF16	RTX 5090	38	256/8	70.2	212	uniform	33.2
	GPT-OSS-120B-MXFP4	RTX PRO 6000 Blackwell	57	128/4	139.2	378	trace	36.8
	DeepSeek-V4-Flash	RTX PRO 6000 Blackwell	115	256/6	57.4	183	uniform	31.3
	Qwen3-30B-A3B-Instruct-2507	RTX 5090	0	128/8	25.7	199	no slots	12.9
Pipelined sharding (Ukarande et al., 2026)	Qwen3-30B-A3B-Instruct-2507	RTX 5090	44	128/8	32.1	1032	trace	3.1
	Qwen3-235B-A22B-Instruct-2507	RTX 5090	0	128/8	7.7	40	no slots	19.3
	Qwen3-235B-A22B-Instruct-2507	RTX 5090	42	128/8	11.5	108	uniform	10.7
	Qwen3-30B-A3B-Instruct-2507	RTX 5070 Ti	107	128/8	54.9	511	trace	10.8
	Laguna S 2.1	RTX 5090	66	256/10	31.3	88	uniform	35.5
llama.cpp fork (leloch) (leloch, 2026a)	Qwen3-30B-A3B Q4_K_M derivative	Radeon RX 7600 8GB	32	128/8	16.5	169	trace	9.8

O ARTIFACT DESCRIPTION

- **System.** A patch to llama.cpp commit 4da6337 (the expert cache, FETCH, the mailbox and the llama-ec-bench instrumentation: teacher-forced loss, logit dumps, routing lookahead records); built with CUDA 13.0 for compute capability 12.0.
- **Jobs.** One shell script per job; each installs FreeToken at commit 0d652e7, downloads the models from Hugging Face, runs the bandwidth probes, writes any on-machine prediction, runs the client, and stores raw rows (one JSON line per problem), server logs and engine counters. A small runner rents a Vast instance, streams a heartbeat, fetches the results and only then destroys the instance.
- **Analysis.** scripts/*_stats.py compute every interval from the raw rows; scripts/perlayer_model.py, speed_limit_v2.py, shapley_gap.py, policy_study.py, foresight_exact.py, batch_k_trace.py, fig_audit_sol.py and scorecard.py produce the model comparison, the limit, the accounting, the trace studies, the audit figure and the scorecard; scripts/wsg_tables.py and wsg_numbers2.py generate every table, figure and number in this paper from their outputs. The audit’s 52 published rows and the physical speed-of-light model they are scored against are data/published_measurements.json, data/audit/ and scripts/audit.py.

- **Cost.** The 124 rentals (74 distinct offers) of jobs 058–108 cost 87.2 US dollars at their hourly GPU prices, by the ledger in the repository; storage and download charges, which the ledger omits, add to this.